RINGS AND MODULES

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Chapter 1

RINGS

January 19th.

Of course, we begin with the definition of a ring.

Definition 1.1. A *ring* is a triple $(R, +, \cdot)$ where R is a set, and $+$ and $\cdot$ are binary operations on R such that the following axioms are satisfied:

- $(R, +)$ is an abelian group. The identity element of this group is denoted by 0_R, and the (additive) inverse of an element $a \in R$ is denoted by $-a$.
- The property of *associativity* of $\cdot$ holds; i.e., for all $a, b, c \in R$, we have $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- The property of *distributivity* of $\cdot$ over $+$ holds; i.e., for all $a, b, c \in R$, we have

$$a \cdot (b + c) = a \cdot b + a \cdot c, \tag{1.1}$$
$$(a + b) \cdot c = a \cdot c + b \cdot c. \tag{1.2}$$

Rings may be written simply as R instead of the triple. The ring R is termed a *ring with unity* if there exists an element $1_R \in R$ such that for all $a \in R$, we have $1_R \cdot a = a \cdot 1_R = a$. Some examples of rings with unity include $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $M_n(\mathbb{R})$ with the usual addition and multiplication. A ring R is said to be a *commutative ring* if for all $a, b \in R$, we have $a \cdot b = b \cdot a$. Examples of commutative rings include $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, but $M_n(\mathbb{R})$ is not commutative for $n \geq 2$. Lastly, a commutative ring R with unity is termed a *field* if every non-zero element of R has a multiplicative inverse; i.e., for every $a \in R \setminus \{0_R\}$, there exists an element $b \in R$ such that $a \cdot b = b \cdot a = 1_R$. Examples of fields include $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, but $\mathbb{Z}$ is not a field.

Example of rings without unity include $2\mathbb{Z}$ with the usual addition and multiplication, and the set of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ that vanish at 0, with the usual addition and multiplication of functions. Another class of rings we previously studied was $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 2$, with the usual addition and multiplication modulo n. This ring has unity, but is a field if and only if n is prime.

Definition 1.2. Let R be a ring with unity. An element $a \in R$ is called a *unit* if there exists an element $b \in R$ such that $a \cdot b = b \cdot a = 1_R$.

For example, in the ring $\mathbb{Z}/n\mathbb{Z}$, an element $\bar{a}$ is a unit if and only if $\gcd(a, n) = 1$. The set of all units in a ring R with unity is denoted by $R^\times$. It can be easily verified that $(R^\times, \cdot)$ is an abelian group.

1.1 Properties and Maps

Some basic properties may be inferred.

Proposition 1.3. *Let R be a ring with unity. Then,*

- *1_R is the unique multiplicative identity in R.*
- *$1_R \cdot 0_R = 0_R$. In general, $a \cdot 0_R = 0_R$ for all $a \in R$.*
- *$-1_R \cdot a = -a$ for all $a \in R$.*

Proof.
- This is left as an exercise to the reader.
- $1_R \cdot 0_R = 1_R$ is trivial since 1_R is the multiplicative identity. For the general case, let $a \in R$. Then,

$$a \cdot 0_R = a \cdot (0_R + 0_R) = a \cdot 0_R + a \cdot 0_R \implies a \cdot 0_R = 0_R \tag{1.3}$$

by the addition of $-(a \cdot 0_R)$ on both sides.

- Let $a \in R$. Then,

$$(-1_R \cdot a) + a = (-1_R + 1_R) \cdot a = 0_R \implies -1_R \cdot a = -a. \tag{1.4}$$

■

The subscript R in 0_R and 1_R may be dropped when the context is clear. We move on to some special maps.

Definition 1.4. A *ring homomorphism* is a map $\varphi : (R, +, \cdot) \to (S, \oplus, \odot)$ between two rings such that for all $a, b \in R$, we have

$$\varphi(a + b) = \varphi(a) \oplus \varphi(b), \quad \varphi(a \cdot b) = \varphi(a) \odot \varphi(b). \tag{1.5}$$

Most of the time, we shall drop $\oplus$ and $\odot$ when the context is clear. Some examples of ring homomorphisms include the map $\varphi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ defined by $\varphi(a) = \bar{a}$ for all $a \in \mathbb{Z}$, and the inclusion map from $\mathbb{Z}$ to $\mathbb{Q}$. Non-examples include $n \mapsto -n$ from $\mathbb{Z}$ to $\mathbb{Z}$, and the determinant map from $M_n(\mathbb{R})$ to $\mathbb{R}$.

Let $(\mathbb{Z} \times \mathbb{Z}, +, \cdot)$ be the ring where addition and multiplication are defined component-wise. Then the map $Z \to \mathbb{Z} \times \mathbb{Z}$ defined by $a \mapsto (a, 0)$ is a ring homomorphism since it preserves both addition and multiplication. However, the unity of $\mathbb{Z}$ is mapped to $(1, 0)$, which is not the unity of $\mathbb{Z} \times \mathbb{Z}$. Thus, ring homomorphisms need not map unity to unity.

Definition 1.5. Let R be a ring with $S \subseteq R$ a subset. Then, S is called a *subring* of R if $(S, +, \cdot)$ is itself a ring with the operations inherited from R.

Again, even if R has unity, a subring S need not have the same unity as R or even a unity at all.

January 23rd.

Definition 1.6. A ring homomorphism $\varphi : R \to S$ is termed a *ring monomorphism* if it is injective, a *ring epimorphism* if it is surjective, and a *ring isomorphism* if it is bijective. If there exists a ring isomorphism from R to S, then R and S are said to be *isomorphic*, denoted by $R \cong S$.

Note that if $\varphi : R \to S$ is bijective, then its inverse $\varphi^{-1} : S \to R$ is a ring homomorphism. We look at some examples of rings and mappings.

Example 1.7. Let X be any set and let $R := \{f : X \to \mathbb{R}\}$ be the set of all functions from X to $\mathbb{R}$. Then, $(R, +, \cdot)$ is a ring where addition and multiplication are defined pointwise; i.e., for all $f, g \in R$ and $x \in X$, $(f + g)(x) := f(x) + g(x)$ and $(f \cdot g)(x) := f(x) \cdot g(x)$. The additive identity is the zero function $0 : X \to \mathbb{R}$ defined by $0(x) = 0$ for all $x \in X$, and the multiplicative identity is the constant function $1 : X \to \mathbb{R}$ defined by $1(x) = 1$ for all $x \in X$. It is easy to verify that all ring axioms are

satisfied. Moreover, this ring is commutative and has unity. Note that $\mathbb{R}$ can be replaced by any ring S to form the ring of functions from X to S. In such a case, R is a (commutative) ring with unity if and only if S is a (commutative) ring with unity.
In the special case that $X = \{1, 2, \ldots, n\}$ for some $n \in \mathbb{N}$, the ring R is isomorphic to the ring $(\mathbb{R}^n, +, \cdot)$ where addition and multiplication are defined component-wise. The isomorphism $\varphi : R \to \mathbb{R}^n$ is given by $\varphi(f) = (f(1), f(2), \ldots, f(n))$ for all $f \in R$.

Example 1.8. Continuing from the previous example, let $X = [a, b]$. Note that the R in this case is the set of all functions from the interval $[a, b]$ to $\mathbb{R}$, which is not a very manageable set. Thus, we may consider the subset $C([a, b], \mathbb{R}) \subseteq R$ consisting of all continuous functions from $[a, b]$ to $\mathbb{R}$. It is easy to verify that $C([a, b], \mathbb{R})$ is a subring of R. Similarly, one defined $C^n([a, b], \mathbb{R})$ to be the set of all n-times continuously differentiable functions from $[a, b]$ to $\mathbb{R}$, and $C^\infty([a, b], \mathbb{R})$ to be the set of all infinitely differentiable functions from $[a, b]$ to $\mathbb{R}$. Both of these are subrings of R as well.

Example 1.9. The set $\mathbb{Z}[i] := \{a + bi : a, b \in \mathbb{Z}\}$ is a subring of the field $\mathbb{C}$. It is easy to verify that $\mathbb{Z}[i]$ is a ring with unity, but it is not a field since, for example, the element $1 + i$ does not have a multiplicative inverse in $\mathbb{Z}[i]$. Note that there is a natural bijection $\varphi : \mathbb{Z}[i] \to \mathbb{Z}^2$ defined by $\varphi(a + bi) = (a, b)$ for all $a + bi \in \mathbb{Z}[i]$, where $\mathbb{Z}^2$ has component-wise addition and multiplication. However, this map is not a ring isomorphism since it does not preserve multiplication; for example, $\varphi(i \cdot i) = \varphi(-1) = (-1, 0)$, but $\varphi(i) \cdot \varphi(i) = (0, 1) \cdot (0, 1) = (0, 1)$.

1.1.1 Polynomials

Let R be a ring. The polynomial ring in the variable x with coefficients from R is defined as follows:

Definition 1.10. The *polynomial ring* $R[x]$ is defined as

$$R[x] := \{f : \mathbb{N}_0 \to R \mid f(n) = 0 \text{ for all but finitely many } n \in \mathbb{N}_0\}. \tag{1.6}$$

The elements of $R[x]$ are called *polynomials* in the variable x with coefficients from R. For $f, g \in R[x]$ and $n \in \mathbb{N}_0$, addition is defined as

$$(f + g)(n) := f(n) + g(n) \quad \text{for all } n \in \mathbb{N}_0, \tag{1.7}$$

and multiplication is defined as

$$(f \cdot g)(n) := \sum_{k=0}^{n} f(k) \cdot g(n - k) \quad \text{for all } n \in \mathbb{N}_0. \tag{1.8}$$

Alternatively, a polynomial $f \in R[x]$ may be expressed in the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n, \tag{1.9}$$

where $a_i = f(i)$ for all $0 \leq i \leq n$ and $f(k) = 0$ for all $k > n$. For $0 \neq f \in R[x]$ as above with $a_n \neq 0_R$, the integer n is called the *degree* of f, denoted by $\deg(f)$. The degree of the zero polynomial is usually left undefined, or changed upon convention. Also note that $f \cdot g \in R[x]$ since $f \cdot g(k) = 0_R$ for all $k > \deg(f) + \deg(g)$.

Proposition 1.11. *For a ring R, the polynomial ring $R[x]$ is, indeed, a ring with unity under the operations defined above. If R is commutative, then so is $R[x]$. The map $\iota : R \to R[x]$ defined by $\iota(a) = f_a$ where $f_a(0) = a$ and $f_a(n) = 0_R$ for all $n \geq 1$ is a ring monomorphism.*

Proof. That $(R[x], +)$ forms an abelian group is clear. The associativity of multiplication is verified as

follows: let $f, g, h \in R[x]$ and $n \in \mathbb{N}_0$. Then,

$$\begin{aligned}((f \cdot g) \cdot h)(n) &= \sum_{k=0}^{n} (f \cdot g)(k) \cdot h(n-k) = \sum_{k=0}^{n} \left(\sum_{j=0}^{k} f(j) \cdot g(k-j) \right) \cdot h(n-k) \\ &= \sum_{j=0}^{n} f(j) \cdot \left(\sum_{k=j}^{n} g(k-j) \cdot h(n-k) \right) = \sum_{j=0}^{n} f(j) \cdot (g \cdot h)(n-j) = (f \cdot (g \cdot h))(n). \quad (1.10)\end{aligned}$$

The distributive properties follow similarly. The unity in $R[x]$ is the polynomial $1_{R[x]}$ defined by $1_{R[x]}(0) = 1_R$ and $1_{R[x]}(n) = 0_R$ for all $n \geq 1$. Finally, it is easy to verify that ι is a ring homomorphism, and it is injective since $\iota(a) = \iota(b)$ implies that $a = b$. ■

With $R[x]$ established as a ring, we may consider a higher level of abstraction, by considering polynomials over this polynomial ring itself; that is, $(R[x])[y]$. Elements of this ring look like

$$f(x, y) = a_{00} + a_{10}x + a_{01}y + a_{20}x^2 + a_{11}xy + a_{02}y^2 + \cdots + a_{mn}x^m y^n, \quad (1.11)$$

where $a_{ij} \in R$ for all $i, j \geq 0$ and $a_{ij} = 0_R$ for all but finitely many pairs (i, j). We have already shown that $R[x]$ is a ring, so it follows that $(R[x])[y]$ is also a ring. This ring is usually denoted by $R[x, y]$. For $f \in R[x, y]$ as above with $a_{mn} \neq 0_R$, the degree of f is defined as $\deg(f) = m + n$. Similarly, one may define $R[x_1, x_2, \ldots, x_n]$ for any $n \in \mathbb{N}$. For a countable number of indeterminates, one may define $R[x_1, x_2, x_3, \ldots]$ as the union $\bigcup_{n=1}^{\infty} R[x_1, x_2, \ldots, x_n]$.

Example 1.12. Let $e \in \mathbb{R}$ be the Euler's number (or any transcendental number). Then $\mathbb{Z}[e] \subseteq \mathbb{C}$ is the smallest subring of $\mathbb{C}$ containing both $\mathbb{Z}$ and e. Here, $\mathbb{Z}[e]$ consists of all polynomials in e with integer coefficients; i.e., all elements of the form $a_0 + a_1 e + a_2 e^2 + \cdots + a_n e^n$ where $n \geq 0$ and $a_i \in \mathbb{Z}$. Since e is transcendental, there are no non-trivial polynomial relations among the powers of e with integer coefficients. Thus, the map $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[e]$ defined by $\varphi(f) = f(e)$ for all $f \in \mathbb{Z}[x]$ is a ring isomorphism.

1.2 Ideals

Definition 1.13. Let R be a commutative ring with unity. A subset $I \subseteq R$ is called an *ideal* of R if the following conditions hold:

- for all $a, b \in I$, we have $a + b \in I$,
- for all $a \in I$ and $r \in R$, we have $r \cdot a \in I$.

Note that the first condition implies that $(I, +)$ is a subgroup of $(R, +)$. Some examples of ideals include the set $\{0_R\}$, the ring R itself, and the set $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for any $n \in \mathbb{Z}_{\geq 0}$ as an ideal of the ring $\mathbb{Z}$. A non-example is $\mathbb{Z}$ in $\mathbb{R}$; it is a subring, but not an ideal since, for example, $1 \in \mathbb{Z}$ but $\pi \cdot 1 = \pi \notin \mathbb{Z}$. Note that if $1_R \in I$, then $I = R$.

Example 1.14. Let us look at ideals of $\mathbb{R}$. Trivially, $\{0\}$ and $\mathbb{R}$ are ideals of $\mathbb{R}$. We claim that these are the only ideals of $\mathbb{R}$. To see this, let I be any ideal of $\mathbb{R}$ such that $I \neq \{0\}$. Then, there exists some $a \in I$ such that $a \neq 0$. Since $\mathbb{R}$ is a field, a has a multiplicative inverse $a^{-1} \in \mathbb{R}$. Thus, $1 = a^{-1} \cdot a \in I$, which implies that $I = \mathbb{R}$. In fact, this argument shows that in any field, the only ideals are the zero ideal and the field itself.

Example 1.15. We examine ideals of $\mathbb{Z}$. From group theory, we know that every subgroup of $(\mathbb{Z}, +)$ is of the form $n\mathbb{Z}$ for some $n \in \mathbb{Z}_{\geq 0}$, so $n\mathbb{Z}$ are the only candidates for ideals of $\mathbb{Z}$. In fact, each $n\mathbb{Z}$ is an ideal of $\mathbb{Z}$ since for all $a, b \in n\mathbb{Z}$, we have $a + b \in n\mathbb{Z}$, and for all $a \in n\mathbb{Z}$ and $r \in \mathbb{Z}$, we have $r \cdot a \in n\mathbb{Z}$. Thus, the ideals of $\mathbb{Z}$ are precisely the sets $n\mathbb{Z}$ for $n \in \mathbb{Z}_{\geq 0}$, and $\mathbb{Z}$.

Proposition 1.16. *Let $f : R \to S$ be a ring homomorphism between two commutative rings with unity. Then, the* kernel *of f, defined as*

$$\ker f := \{a \in R \mid f(a) = 0_S\}, \tag{1.12}$$

is an ideal of R. Moreover, f is a ring monomorphism if and only if $\ker f = \{0_R\}$.

Proof. Let $a, b \in \ker f$ and $r \in R$. Then,

$$f(a+b) = f(a) + f(b) = 0_S + 0_S = 0_S, \tag{1.13}$$

so $a + b \in \ker f$. Also,

$$f(r \cdot a) = f(r) \cdot f(a) = f(r) \cdot 0_S = 0_S, \tag{1.14}$$

so $r \cdot a \in \ker f$. Thus, $\ker f$ is an ideal of R.

Now, suppose that f is a ring monomorphism. Let $a \in \ker f$. Then, $f(a) = 0_S = f(0_R)$. Since f is injective, we have $a = 0_R$, so $\ker f = \{0_R\}$. Conversely, suppose that $\ker f = \{0_R\}$. Let $a, b \in R$ such that $f(a) = f(b)$. Then,

$$f(a-b) = f(a) - f(b) = 0_S, \tag{1.15}$$

so $a - b \in \ker f$. Thus, $a - b = 0_R$, which implies that $a = b$. Therefore, f is injective. ■

January 24th.

Let R be a ring with unity and R_i be a colelction of subrings of R containing the nuity. Then $\bigcap_i R_i$ is also a subring of R containing the unity. If I_j is a collection of ideals of R, then $\bigcap_j I_j$ is also an ideal of R. Thus, given any subset $S \subseteq R$, we may define the ideal generated.

Definition 1.17. Let R be a commutative ring with unity and $I \subseteq R$ be an ideal. Let $S \subseteq I$ be a set. We say S is a *generating set* of I if I is the smallest ideal containing S.

Proposition 1.18. *Let R be a commutative ring with unity and $S \subseteq R$ be any subset. Then, the ideal generated by S, denoted by (S), is given by*

$$(S) = \left\{ \sum_{i=1}^{n} r_i s_i : n \geq 0, r_i \in R, s_i \in S \text{ for all } 1 \leq i \leq n \right\}. \tag{1.16}$$

Proof. Let $S \subseteq I$, a subset of an ideal. We claim that $(S) \subseteq I$. Let $\alpha \in I$. THen, $\alpha = r_1 x_1 + \cdots + r_n x_n$ for some $n \geq 0$, $r_i \in R$ and $x_i \in S$ for all $1 \leq i \leq n$. Since I is an ideal, we have $r_i x_i \in I$ for all $1 \leq i \leq n$, and thus $\alpha \in I$. Therefore, $(S) \subseteq I$. ■

With this, we introduce the notation that if $\{x_1, \ldots, x_n\} \subseteq R$, then $I = (x_1, \ldots, x_n) = Rx_1 + \cdots + Rx_n$ is the ideal generated by $x_1, \ldots, x_n$. Let us look at some exmaples.

Example 1.19. In the ring $\mathbb{Z}$, $(2, 3) = \mathbb{Z}$ since $1 = 3 - 1 \cdot 2 \in (2, 3)$. More generally, for any $a, b \in \mathbb{Z}$, we have $(a, b) = \mathbb{Z}$ if and only if $\gcd(a, b) = 1$. Moreover, in $\mathbb{Z}$, every ideal can be generated by a single element; i.e., every ideal is of the form (n) for some $n \in \mathbb{Z}_{\geq 0}$.

Example 1.20. In $\mathbb{Z}[x]$, the ideal $(2, x)$ consists of all polynomials with integer coefficients where the constant term is even. That is, $(2, x) = 2\mathbb{Z}[x] + x\mathbb{Z}[x]$.

Also note that a union of ideals need not be an ideal. For example, in $\mathbb{Z}$, the sets $2\mathbb{Z}$ and $3\mathbb{Z}$ are ideals, but their union $2\mathbb{Z} \cup 3\mathbb{Z}$ is not an ideal. This, however, calls for a more general construction.

Definition 1.21. Let R be a commutative ring with unity. If I_1, I_2 are two ideals, we then define their sum as $I_1 + I_2 = (I_1 \cup I_2)$.

It is easy to verify that $I_1 + I_2 = \{a + b \mid a \in I_1, b \in I_2\}$. This definition may be extended to a finite number of ideals in the obvious way.

1.3 Other Rings

Definition 1.22. Let G be a group and k be a field. We define $R[G]$ to be the set of all functions $f : G \to k$ such that $f(x) = 0$ for all but finitely many $x \in G$. Addition is defined pointwise as

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in G, \tag{1.17}$$

and multiplication is defined as

$$(f \cdot g)(x) = \sum_{yz=x} f(y)g(z) = \sum_{y \in G} f(xy^{-1})g(y) \quad \text{for all } x \in G. \tag{1.18}$$

The ring $R[G]$ is called the *group ring* of G over k.

If G is a finite group with $G = \{e, x_2, \ldots, x_n\}$ then $R[G] = \{a_1 e + a_2 x_2 + \cdots + a_n x_n \mid a_i \in \mathbb{C}\}$. Verify that $R[G]$ is a ring with unity under the operations defined above. This ring, however, may not be commutative.

Definition 1.23. Let R be a ring and $x \in R$. x is termed *nilpotent* if there exists some $n \in \mathbb{N}$ such that $x^n = 0$. If R is commutative, $x \in R$ is called a *zero divisor* if there exists some $y \in R \setminus \{0\}$ such that $x \cdot y = 0$.

Note that nilpotents are zero divisors in a commutative ring, but the converse need not be true. For example, in the ring $\mathbb{Z}/6\mathbb{Z}$, the element $\bar{2}$ is a zero divisor since $\bar{2} \cdot \bar{3} = \bar{0}$, but it is not nilpotent since $\bar{2}^n \neq \bar{0}$ for all $n \geq 1$.

Definition 1.24. A commutative ring with unity R is called a *reduced ring* if it has no non-zero nilpotent elements. It is called an *integral domain* if it has no non-zero zero divisors.

Proposition 1.25. *Let R be an integral domain. Then, if $x, y \in R$ are such that $x \cdot y = 0$, then either $x = 0$ or $y = 0$.*

Proof. If $x \neq 0$, then since R is an integral domain, x is not a zero divisor. Thus, y must be 0. Similarly, if $y \neq 0$, then x must be 0. ■

Proposition 1.26. *Every integral domain is a reduced ring.*

Proof. Let R be an integral domain and let $x \in R$ be nilpotent. Then, there exists some $n \in \mathbb{N}$ such that $x^n = 0$. If $x \neq 0$, then since R is an integral domain, x is not a zero divisor. However, this contradicts the fact that $x^n = 0$. Thus, we must have $x = 0$, so R has no non-zero nilpotent elements. ■

January 29th.

In an integral domain R, if $ab = ac$ for some $a, b, c \in R$, then either $a = 0$ or $b = c$. Let us look at some examples of integral domains.

Example 1.27. The ring $\mathbb{Z}$ is an integral domain since it has no non-zero zero divisors. Similarly, the rings $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are integral domains as well. More generally, any field is an integral domain. Moreover, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

Example 1.28. Note that R is an integral domain if and only if $R[x]$ is an integral domain.

Another small result is as follows: if R is an integral domain and R' is a subring of R containing the unity, then R' is also an integral domain. Some non-examples of integral domains include $\mathbb{Z}^2$, $C[0,1]$, $C^\infty[0,1]$, etc.

1.4 Quotient Rings and Isomorphism Theorems

From here on, we shall assume that all rings are commutative with unity unless otherwise stated.

Definition 1.29. Let R be a ring and I be an ideal of R. The *quotient ring* R/I is defined as the set of all cosets of I in R; i.e.,

$$R/I := \{a + I : a \in R\}. \tag{1.19}$$

Addition and multiplication in R/I are defined as follows: for all $a, b \in R$,

$$(a + I) + (b + I) := (a + b) + I, \tag{1.20}$$
$$(a + I) \cdot (b + I) := (a \cdot b) + I. \tag{1.21}$$

Of course, we must verify that these operations are well-defined. Note that I is a normal subgroup of $(R, +)$ since R is abelian under addition, so R/I is an abelian group under addition. We verify that multiplication is well-defined as follows: let $a, a', b, b' \in R$ such that $a + I = a' + I$ and $b + I = b' + I$. Then, there exist $i_1, i_2 \in I$ such that $a' = a + i_1$ and $b' = b + i_2$. Thus,

$$\begin{aligned}(a' \cdot b') + I &= ((a + i_1) \cdot (b + i_2)) + I = (a \cdot b + a \cdot i_2 + i_1 \cdot b + i_1 \cdot i_2) + I \\ &= (a \cdot b) + I,\end{aligned} \tag{1.22}$$

since $a \cdot i_2, i_1 \cdot b, i_1 \cdot i_2 \in I$. Therefore, multiplication is well-defined. Moreover, it is easy to verify that R/I is a ring with unity under these operations, where the additive identity is $0 + I$ and the multiplicative identity is $1 + I$. One also has the *quotient map* naturally defined as

$$q : R \to R/I, \quad q(a) = a + I \quad \text{for all } a \in R. \tag{1.23}$$

It is easy to verify that q is a ring epimorphism with kernel I. The most common example of a quotient ring is $\mathbb{Z}/n\mathbb{Z}$, which is isomorphic to the quotient ring $\mathbb{Z}/(n\mathbb{Z})$.

Example 1.30. In the ring $\mathbb{Q}[x]$, let $I = (x^2 - 2) = (x^2 - 2)\mathbb{Q}$. Then, the quotient $\mathbb{Q}[x]/(x^2 - 2)$ is indeed a quotient ring. It is also a field, and may be written as $\mathbb{Q}[\sqrt{2}]$. However, the quotient ring $\mathbb{R}[x]/(x^2 - 2)$ is not an integral domain since $(x - \sqrt{2} + I)(x + \sqrt{2} + I) = x^2 - 2 + I = I$.

We are now fit to show the isomorphism theorems for rings.

Theorem 1.31 (The *first isomorphism theorem* for rings)**.** *Let $\varphi : R \to S$ be a ring homomorphism. Let $I = \ker \varphi$. Then there exists a unique ring monomorphism $\overline{\varphi} : R/I \to S$ such that $\varphi = \overline{\varphi} \circ q$, where $q : R \to R/I$ is the quotient map. Moreover, if φ is surjective, then $\overline{\varphi}$ is a ring isomorphism.*

Proof. Define the map $\overline{\varphi} : R/I \to S$ as follows: for all $a + I \in R/I$, let

$$\overline{\varphi}(a + I) = \varphi(a). \tag{1.24}$$

We must verify that this map is well-defined. Let $a, b \in R$ such that $a + I = b + I$. Then, there exists some $i \in I$ such that $b = a + i$. Thus,

$$\varphi(b) = \varphi(a + i) = \varphi(a) + \varphi(i) = \varphi(a) + 0_S = \varphi(a), \tag{1.25}$$

so $\overline{\varphi}$ is well-defined. It is easy to verify that $\overline{\varphi}$ is a ring homomorphism. Also, for all $a \in R$,

$$(\overline{\varphi} \circ q)(a) = \overline{\varphi}(a + I) = \varphi(a), \tag{1.26}$$

so $\varphi = \overline{\varphi} \circ q$.

Now, suppose that $\overline{\varphi}(a + I) = 0_S$ for some $a + I \in R/I$. Then, $\varphi(a) = 0_S$, so $a \in I$. Thus, $a + I = I$, which is the additive identity in R/I. Therefore, $\overline{\varphi}$ is injective.

Finally, if φ is surjective, then for any $s \in S$, there exists some $a \in R$ such that $\varphi(a) = s$. Thus,

$$\overline{\varphi}(a + I) = \varphi(a) = s, \tag{1.27}$$

so $\overline{\varphi}$ is surjective as well. Therefore, $\overline{\varphi}$ is a ring isomorphism. ■

Proposition 1.32. *Let R be a ring and I be an ideal of R. Then there is a bijection between the set of all ideals of R containing I and the set of all ideals of the quotient ring R/I.*

Proof. We make use of the quotient map $q : R \to R/I$. Let J be an ideal of R such that $I \subseteq J$. The bijection is given by sending J to $J/I := q(J) = \{a + I : a \in J\}$, and sending K, an ideal of R/I, to $q^{-1}(K) = \{a \in R : q(a) \in K\}$. We first show that $q(J) = J/I$ is indeed an ideal of R/I, for J an ideal of R containing I. Let $x + I \in J/I$ and $r + I \in R/I$. Then, $(r + I)(x + I) = (r \cdot x) + I$. Since $x \in J$ and J is an ideal of R, we have $r \cdot x \in J$, so $(r + I)(x + I) \in J/I$. Also note that for all $x + I, y + I \in J/I$, we have $(x + I) + (y + I) = (x + y) + I \in J/I$ since $x, y \in J$ and J is an ideal of R. Thus, J/I is an ideal of R/I.

On the other hand, we show that $q^{-1}(K)$ is an ideal of R for K an ideal of R/I. Let $x, y \in q^{-1}(K)$. Then, $q(x), q(y) \in K$, so $q(x + y) = q(x) + q(y) \in K$ since K is an ideal of R/I. Thus, $x + y \in q^{-1}(K)$. Also, for any $r \in R$ and $x \in q^{-1}(K)$, we have $q(r), q(x) \in R/I$ and $q(x) \in K$, so $q(r \cdot x) = q(r) \cdot q(x) \in K$ since K is an ideal of R/I. Thus, $r \cdot x \in q^{-1}(K)$. Therefore, $q^{-1}(K)$ is an ideal of R. Also, if $x \in I$, then $q(x) = x + I = I$, which is the additive identity in R/I and thus belongs to every ideal of R/I. Therefore, $I \subseteq q^{-1}(K)$.

To show that the maps are inverses of each other is left as an exercise. ■

Theorem 1.33 (The *second isomorphism theorem* for rings)**.** *Let R be a ring, and let $S \subseteq R$ be a subring containing the unity. Let I be an ideal of R. Then, $S + I = \{s + i : s \in S, i \in I\}$ is a subring of R containing the unity, $S \cap I$ is an ideal of S, and there is a ring isomorphism*

$$(S + I)/I \cong S/(S \cap I). \tag{1.28}$$

Proof. Let $\alpha, \beta \in S + I$. Then $\alpha = s + x$ and $\beta = s' + y$ for some $s, s' \in S$ and $x, y \in I$. Thus, $\alpha + \beta = (s + s') + (x + y) \in S + I$ since S is a subring and I is an ideal. Also, $\alpha \cdot \beta = (s + x)(s' + y) = ss' + sy + xs' + xy \in S + I$ since $ss' \in S$, $sy, xs', xy \in I$. Therefore, $S + I$ is a subring of R containing the unity.

Note that the inclusion map $i : S \to R$ is a ring homomorphism. Thus, by the proposition above, $S \cap I = i^{-1}(I)$ is an ideal of S. Also, $I \subseteq S + I$ is an ideal of $S + I$. Now let $\varphi : S \to (S + I)/I$ be the map $\varphi = q \circ i$, where $q : S + I \to (S + I)/I$ is the quotient map. It is easy to verify that φ is a ring homomorphism with kernel

$$\ker \varphi = \{a \in S \mid q \circ i(a) = I\} = \{a \in S \mid a + I = I\} = S \cap I. \tag{1.29}$$

Moreover, φ is surjective since for any $s + i + I \in (S + I)/I$ where $s \in S$ and $i \in I$, we have $\varphi(s) = s + I = s + i + I$. Thus, by the first isomorphism theorem, we have the desired isomorphism. ■

January 30th.

Theorem 1.34 (The *third isomorphism theorem* for rings)**.** *Let R be a ring, and let $J \subseteq I$ be two ideals of R. Then, $I/J = \{a + J : a \in I\}$ is an ideal of the quotient ring R/J, and there is a ring isomorphism*

$$(R/J)/(I/J) \cong R/I. \tag{1.30}$$

Proof. Let $q_R : R \to R/J$ be the quotient map, and $q_{R/J} : R/J \to (R/J)/(I/J)$ be the quotient map. Thus, the composition $\varphi = q_{R/J} \circ q_R : R \to (R/J)/(I/J)$ is a surjective ring homomorphism. The kernel of φ is given by

$$\ker \varphi = \{x \in R \mid \varphi(x) = J + I/J\} = \{x \in R \mid q_R(x) \in I/J\} = \{x \in R \mid x + J \in I/J\} = I. \tag{1.31}$$

Thus, by the first isomorphism theorem, we have the desired isomorphism. ■

We look at some applications of the isomorphism theorems.

Example 1.35. Let $I = (5) \subseteq \mathbb{Z}[x]$. We claim that $\mathbb{Z}[x]/5\mathbb{Z}[x] \cong (\mathbb{Z}/5\mathbb{Z})[x]$. To see this, we make use of the first isomorphism theorem. Let $\varphi : \mathbb{Z}[x] \to (\mathbb{Z}/5\mathbb{Z})[x]$ be the map defined by

$$\varphi\left(a_0 + a_1x + a_2x^2 + \cdots + a_nx^n\right) = \bar{a}_0 + \bar{a}_1x + \bar{a}_2x^2 + \cdots + \bar{a}_nx^n, \tag{1.32}$$

where $\bar{a}_i$ is the image of a_i in $\mathbb{Z}/5\mathbb{Z}$ for all $0 \leq i \leq n$. It is easy to verify that φ is a surjective ring homomorphism with kernel $5\mathbb{Z}[x]$. Thus, by the first isomorphism theorem, we have the desired isomorphism.

Example 1.36. Let $(x) \subseteq \mathbb{Z}[x]$. We claim that $\mathbb{Z}[x]/(x) \cong \mathbb{Z}$. To see this, we make use of the second isomorphism theorem. Let $S = \mathbb{Z} \subseteq \mathbb{Z}[x]$. Then, $S + (x) = \mathbb{Z}[x]$ since for any $f(x) \in \mathbb{Z}[x]$, we have $f(x) = f(0) + (f(x) - f(0)) \in S + (x)$. Also, $S \cap (x) = \{0\}$ since the only constant polynomial in (x) is the zero polynomial. Thus, by the second isomorphism theorem, we have

$$\mathbb{Z}[x]/(x) \cong S/(S \cap (x)) = S/\{0\} \cong \mathbb{Z}. \tag{1.33}$$

Example 1.37. Again, let $I = (x^2 - 4, 2) \subseteq \mathbb{Z}[x]$. We claim the isomorphism

$$\mathbb{Z}[x]/(x^2 - 4, 2) \cong \mathbb{Z}/2\mathbb{Z}[x]/(x^2). \tag{1.34}$$

To see this, we make use of the third isomorphism theorem. Let $J = (2) \subseteq I$. Then, by the third isomorphism theorem, we have

$$\mathbb{Z}[x]/I \cong (\mathbb{Z}[x]/J)/(I/J) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 - 4 + J) = (\mathbb{Z}/2\mathbb{Z})[x]/(x^2), \tag{1.35}$$

since $x^2 - 4 + J = x^2 + J$ in $(\mathbb{Z}/2\mathbb{Z})[x]$.

1.5 Prime and Maximal Ideals

Definition 1.38. Let R be a ring. An ideal $P \subseteq R$ is called a *prime ideal* if $P \neq R$ and for all $a, b \in R$ such that $a \cdot b \in P$, we have either $a \in P$ or $b \in P$.

Of course, the most common example of a prime ideal is (0_R) in an integral domain R. Another example is $(p) = p\mathbb{Z}$ in $\mathbb{Z}$ for any prime p. Note that if R is a field, then the only prime ideal of R is (0_R).

Theorem 1.39. *Let I be an ideal of a ring R. Then, I is a prime ideal if and only if the quotient ring R/I is an integral domain.*

Proof. Suppose that R/I is an integral domain. Let $a, b \in R$ such that $a \cdot b \in I$. Then,

$$(a + I)(b + I) = (a \cdot b) + I = I, \tag{1.36}$$

which is the zero element in R/I. Since R/I is an integral domain, either $a + I = I$ or $b + I = I$, which implies that either $a \in I$ or $b \in I$. Thus, I is a prime ideal. If we now suppose that I is a prime ideal, let $a + I, b + I \in R/I$ such that $(a + I)(b + I) = (a \cdot b) + I = I$. This implies that $a \cdot b \in I$, so either $a \in I$ or $b \in I$. Thus, either $a + I = I$ or $b + I = I$, so R/I is an integral domain. ■

One can also show that there is the natural bijection between ideals of R/I and ideals of R containing I restricts to a bijection between prime ideals of R/I and prime ideals of R containing I.

Example 1.40. We can use this theorem to show (x^2+1) is a prime ideal of $\mathbb{Z}[x]$. Indeed, look at the ring homomorphism $\varphi : \mathbb{Z}[x] \to \mathbb{Z}[i]$ defined by $\varphi(f) = f(i)$ for all $f \in \mathbb{Z}[x]$. For the kernel, let $f \in \ker\varphi$. By the division algorithm, there exist unique $q, r \in \mathbb{Z}[x]$ such that

$$f(x) = (x^2+1)q(x) + r(x), \tag{1.37}$$

where either $r(x) = 0$ or $\deg r < 2$. Plugging in $x = i$ gives $f(i) = 0 = r(i)$. The only way an at most linear polynomial $r(x)$ can be 0 at $x = i$ is if r is the zero polynomial. Hence, $\ker\varphi = (x^2+1)$. Since $\mathbb{Z}[i]$ is an integral domain, by the first isomorphism theorem, we have

$$\mathbb{Z}[x]/(x^2+1) \cong \mathbb{Z}[i] \tag{1.38}$$

showing that (x^2+1) is a prime ideal of $\mathbb{Z}[x]$.

Another notion is the maximal ideal.

Definition 1.41. Let R be a ring. An ideal $M \subseteq R$ is called a *maximal ideal* if $M \neq R$ and there are no ideals I of R such that $M \subsetneq I \subsetneq R$.

That is, if J is an ideal such that $M \subseteq J$, then either $J = M$ or $J = R$. For example, in $\mathbb{Z}$, the ideals $(p) = p\mathbb{Z}$ for prime p are maximal ideals. Note that if R is a field, then the only maximal ideal of R is (0_R). In fact, it may be shown that R is a field if and only if (0) and R are the only ideals.

February 2nd.

Theorem 1.42. *Let I be an ideal of a ring R. Then, I is a maximal ideal if and only if the quotient ring R/I is a field.*

Proof. We work with a set of equivalences. $I \subseteq R$ is a maximal ideal $\iff \{J \mid I \subseteq J \subseteq R\} = \{I, R\}$ $\iff \{K \mid K \text{ is an ideal of } R/I\} = \{I/I, R/I\} \iff R/I$ is a field. ■

A neat corollary of this theorem is that every maximal ideal is a prime ideal. Indeed, if M is a maximal ideal of R, then R/M is a field, and thus an integral domain. Therefore, by the previous theorem, M is a prime ideal. Another theorem guarantees the existence of maximal ideals.

Theorem 1.43. *Every non-zero ring has at least a maximal ideal.*

This theorem, though true in many common cases, requires Zorn's lemma for a general proof.

Proof. Let $(\Omega, \subseteq)$ be the set of all proper ideals of a ring R, ordered by inclusion. Note that $R \supsetneq (0) \in \Omega$, so $\Omega \neq \emptyset$. Zorn's lemma states that if every chain in Ω has an upper bound in Ω, then Ω has a maximal element. Let $\mathcal{C}$ be a chain in Ω. We claim that $I_{\mathcal{C}} = \bigcup_{I \in \mathcal{C}} I$ is an upper bound of $\mathcal{C}$ in Ω. Certainly, for any $I \in \mathcal{C}$, we have $I \subseteq J$. Also, we must verify that $I_{\mathcal{C}}$ is a proper ideal of R.

Let $x, y \in I_{\mathcal{C}}$, and $r \in R$. Then there exist two ideals $I, J \in \mathcal{C}$ such that $x \in I$ and $y \in J$. Since $\mathcal{C}$ is a chain, without loss of generality, suppose that $I \subseteq J$. Thus, $x, y \in J$, so $x + y \in R$ and $r \cdot x \in R$ since J is an ideal of R. Therefore, $I_{\mathcal{C}}$ is an ideal of R. Also, if $I_{\mathcal{C}} = R$, then $1_R \in I_{\mathcal{C}}$, so there exists some ideal $I \in \mathcal{C}$ such that $1_R \in I$. This implies that $I = R$, which contradicts the fact that I is a proper ideal. Thus, $I_{\mathcal{C}}$ is a proper ideal of R, so $I_{\mathcal{C}} \in \Omega$. Therefore, by Zorn's lemma, Ω has a maximal element, which is a maximal ideal of R. ■

Example 1.44. In $\mathbb{C}[x]$, the ideal (x) is maximal since $\mathbb{C}[x]/(x) \cong \mathbb{C}$, which is a field. However, (x^2) is not maximal since $(x^2) \subsetneq (x)$. Another reasoning is that $\mathbb{C}[x]/(x^2)$ is not a field since $(x + (x^2)) \cdot (x + (x^2)) = x^2 + (x^2) = 0 + (x^2)$; it is not an integral domain either, so (x^2) is not even prime. In fact, the maximal ideals of $\mathbb{C}[x]$ are precisely of the form $(x - a)$ for some $a \in \mathbb{C}$.

1.5.1 Jacobson Radical and Nilradical

February 5th.

Definition 1.45. The *Jacobson radical* of a non-zero ring R is defined as

$$\mathrm{Jac}(R) = \bigcap_{\substack{\mathfrak{m} \subseteq R \\ \mathfrak{m} \text{ is a maximal ideal}}} \mathfrak{m}. \tag{1.39}$$

It is the interseciton of ideals, so it is also an ideal. The *nilradical* of R is defined as

$$\mathrm{nil}(R) = \{x \in R \mid x^n = 0 \text{ for some } n \in \mathbb{N}\}. \tag{1.40}$$

The following are some properties of the Jacobson radical and nilradical.

Proposition 1.46. *Let R be a ring. Then,*

1. $\mathrm{nil}(R)$ *is an ideal of* R.
2. $\mathrm{nil}(R) \subseteq \mathrm{Jac}(R)$.
3. $x \in \mathrm{Jac}(R)$ *if and only if* $1 + ax$ *is a unit for all* $a \in R$.
4. $\mathrm{nil}(R) = \bigcap_{\substack{P \subseteq R \\ P \text{ is a prime ideal}}} P$.

Proof. 1. Let $x, y \in \mathrm{nil}(R)$ and $r \in R$. Then, there exist $m, n \in \mathbb{N}$ such that $x^m = 0$ and $y^n = 0$. Thus, $(x+y)^{m+n} = 0$ by the binomial theorem, so $x + y \in \mathrm{nil}(R)$. Also, $(r \cdot x)^m = r^m \cdot x^m = 0$, so $r \cdot x \in \mathrm{nil}(R)$. Therefore, $\mathrm{nil}(R)$ is an ideal of R.

2. Let $x \in \mathrm{nil}(R)$. Then, there exists some $n \in \mathbb{N}$ such that $x^n = 0$. Thus x^n belongs to every ideal of R, so in particular, $x^n \in P$ for every prime ideal and $x^n \in \mathfrak{m}$ for every maximal ideal. However, $x^n \in P$ implies that $x \in P$ since P is prime, and $x^n \in \mathfrak{m}$ implies that $x \in \mathfrak{m}$ since $\mathfrak{m}$ is prime as well. Therefore, x belongs to every prime ideal and every maximal ideal, so $x \in \mathrm{Jac}(R)$. Hence, $\mathrm{nil}(R) \subseteq \mathrm{Jac}(R)$.

3. For the forward implication, let $x \in \mathrm{Jac}(R)$ and $a \in R$. Note that ax cannot be a unit since it is contained in the (maximal) ideals $\mathfrak{m}$. Suppose $1 + ax$ is *not* a unit. Then $I = (1 + ax)$ is a proper ideal of R. If $q : R \to R/I$ is the quotient map, and $\bar{\mathfrak{m}}$ is a maximal ideal of R/I, then we claim that $\mathfrak{m} = q^{-1}(\bar{\mathfrak{m}})$ is a maximal ideal of R containing I. Certainly, $\mathfrak{m}$ is an ideal of R containing I by the proposition above. Also, if there exists some ideal J of R such that $\mathfrak{m} \subsetneq J \subsetneq R$, then $I \subseteq J \subsetneq R$, so $q(J)$ is an ideal of R/I such that $\bar{\mathfrak{m}} \subsetneq q(J) \subsetneq R/I$, contradicting the maximality of $\bar{\mathfrak{m}}$. Thus, $\mathfrak{m}$ is a maximal ideal of R. However, since $x \in \mathrm{Jac}(R)$, we have $x \in \mathfrak{m}$, so $ax \in I \subseteq \mathfrak{m}$. This implies that $1 = (1 + ax) - ax \in \mathfrak{m}$, a contradiction. Therefore, $1 + ax$ is a unit for all $a \in R$.

   For the converse, suppose $1 + ax$ is a unit for all $a \in R$. Let $\mathfrak{m}$ be any maximal ideal of R. If $x \notin \mathfrak{m}$, then the ideal $(\mathfrak{m}, x)$ properly contains $\mathfrak{m}$, so $(\mathfrak{m}, x) = R$. Thus, there exist $m \in \mathfrak{m}$ and $r \in R$ such that $1 = m + r \cdot x$. This implies that $1 - r \cdot x = m \in \mathfrak{m}$, contradicting the fact that $1 - r \cdot x$ is a unit. Therefore, $x \in \mathfrak{m}$. Since $\mathfrak{m}$ was an arbitrary maximal ideal, we have $x \in \mathrm{Jac}(R)$.

4. It is clear that $\mathrm{nil}(R) \subseteq \bigcap_{P \text{ prime}} P$ since maximal ideals are prime ideals. For the converse inclusion, let $x \in \bigcap_{P \text{ prime}} P$. We claim that x is nilpotent. Suppose not. Then, the set $S = \{x^n : n \in \mathbb{N}\}$ does not contain 0. Define the set $\Omega = \{I \subseteq R \mid I \text{ is an ideal and } I \cap S = \emptyset\}$. Inclusion $\subseteq$ is a partial order on Ω. Let $\mathcal{C}$ be a chain in Ω and let $I = \bigcup_{J \in \mathcal{C}} J$. We claim that I is an upper bound of $\mathcal{C}$ in Ω. It is an ideal since if $z, y \in I$, then $z \in J_1$ and $y \in J_2$ with $J_1 \subseteq J_2$. Thus, $z, y \in J_1 \cup J_2$, so $z + y \in J_1 \cup J_2 \subseteq I$. Also, for any $r \in R$ and $z \in I$, we have $z \in J$ for some $J \in \mathcal{C}$, so $r \cdot z \in J \subseteq I$. Therefore, I is an ideal. Moreover, if $J \cap S = \emptyset$ for all $J \in \mathcal{C}$ tells us $I \cap S = \emptyset$. Hence, I is a valid upper bound, and Zorn's lemma guarantess the existence of a maximal P in Ω. We further claim that P is a prime ideal of R. Indeed, $P \subsetneq R$ since $P \cap S = \emptyset$. Let $uv \in P$ for some $u, v \in R$. If both $u, v \notin P$, then the ideals (P, u) and (P, v) properly contain P; that is, $au + y = x^n$ and $bv + z = x^m$ for some $a, b \in R$, $y, z \in P$, and $n, m \in \mathbb{N}$. Thus, $(au + y)(bv + z) = x^{n+m} \in S$, but

$(au+y)(bv+z) = abuv + az + by + yz \in P$ since $uv \in P$ and $y, z \in P$. This contradicts the fact that $P \cap S = \emptyset$. Therefore, either $u \in P$ or $v \in P$, so P is a prime ideal. However, by construction, x is in every prime ideal of R, so $x \in P$, contradicting the fact that $P \cap S = \emptyset$. Hence, x is nilpotent, so $x \in \mathrm{nil}(R)$.

■

February 6th.

Let us look at some examples.

Example 1.47. In $\mathbb{Z}$, $\mathrm{nil}(\mathbb{Z}) = \{0\}$ since the only nilpotent element is 0. Also, $\mathrm{Jac}(\mathbb{Z}) = \{0\}$ since the intersection of all maximal ideals (p) for prime p is $\{0\}$. In the ring $\mathbb{Z}/6\mathbb{Z}$, the nilradical is $\{0\}$ even though it is not an integral domain. The Jacobson radical is $\{0\}$ as well since the maximal ideals are (2) and (3), whose intersection is $\{0\}$.

Example 1.48. For a non-trivial example, let us look at the ring $R = \mathbb{Z}/4\mathbb{Z}$. The nilradical of R is $\{0, 2\}$ since $2^2 = 0$ in R. The Jacobson radical of R is also $\{0, 2\}$ since the only maximal ideal of R is (2). If we take $R = \mathbb{Q}[x, y]/(x^2, y)$, then $\mathrm{nil}(R) = (x)$ since x is nilpotent. The Jacobson radical of R is also (x).

Example 1.49. For an example where the Jacobson radical is strictly larger than the nilradical, consider the power series ring $R = \mathbb{Q}[[x]]$. If $f \in \mathbb{Q}[[x]]$, then f is a unit if and only if the constant term of f is non-zero; if $f = a_0(1-g)$ where $a_0 \in \mathbb{Q}$ and $g \in (x)$, then $f^{-1} = a_0^{-1}(1-g)^{-1}$, where $(1-g)^{-1}$ is given by the geometric series expansion $(1-g)^{-1} = 1 + g + g^2 + \cdots$. Thus, $\mathrm{Jac}(R) = (x)$ since $1 + ax$ is a unit for all $a \in R$. However, $\mathrm{nil}(R) = \{0\}$ since there are no non-zero nilpotent elements in R.

1.6 Product of Rings

Let $R_1, R_2, \ldots, R_n$ be rings. We can define the *product ring* $R_1 \times R_2 \times \cdots \times R_n$ as the set of all n-tuples $(r_1, r_2, \ldots, r_n)$ where $r_i \in R_i$ for all $1 \leq i \leq n$, with addition and multiplication defined componentwise. It is easy to verify that $R_1 \times R_2 \times \cdots \times R_n$ is also a ring, and the unity is given by $(1_{R_1}, 1_{R_2}, \ldots, 1_{R_n})$. The *projection map* $p_i : R_1 \times R_2 \times \cdots \times R_n \to R_i$ is defined by $p_i(r_1, r_2, \ldots, r_n) = r_i$ for all $1 \leq i \leq n$. It is, again, easy to verify that p_i is a surjective ring homomorphism with kernel $R_1 \times \cdots \times R_{i-1} \times \{0_{R_i}\} \times R_{i+1} \times \cdots \times R_n$. Conversely, the *inclusion map* $e_i : R_i \to R_1 \times R_2 \times \cdots \times R_n$ is defined by $e_i(r) = (0_{R_1}, \ldots, 0_{R_{i-1}}, r, 0_{R_{i+1}}, \ldots, 0_{R_n})$ for all $r \in R_i$. It is easy to verify that e_i is an injective ring homomorphism with image $\{0_{R_1}\} \times \cdots \times \{0_{R_{i-1}}\} \times R_i \times \{0_{R_{i+1}}\} \times \cdots \times \{0_{R_n}\}$.

Regarding ideals, we have the following proposition.

Proposition 1.50. *Let $R_1, R_2, \ldots, R_n$ be rings. Then, every ideal of the product ring $R_1 \times R_2 \times \cdots \times R_n$ is of the form $I_1 \times I_2 \times \cdots \times I_n$ where I_i is an ideal of R_i for all $1 \leq i \leq n$.*

Proof. Let I be an ideal of $R_1 \times R_2 \times \cdots \times R_n$. For each $1 \leq i \leq n$, let $I_i = p_i(I)$. We claim that each I_i is an ideal of R_i and $I = I_1 \times I_2 \times \cdots \times I_n$. Let $x_i, y_i \in I_i$ and $r_i \in R_i$. Then, there exist $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in I$ such that $p_i(x) = x_i$ and $p_i(y) = y_i$. Thus, $x + y \in I$ since I is an ideal of the product ring, so $x_i + y_i = p_i(x + y) \in I_i$. Also, $r_i \cdot x_i = p_i(e_i(r_i) \cdot x) \in I_i$ since $e_i(r_i) \cdot x \in I$. Therefore, I_i is an ideal of R_i.

Let use first show that $I \supseteq I_1 \times I_2 \times \cdots \times I_n$. Let $(a_1, a_2, \ldots, a_n) \in I_1 \times I_2 \times \cdots \times I_n$. Let $x^{(i)} \in I$ such that $p_i(x^{(i)}) = a_i$ for all $1 \leq i \leq n$. Then $e_i(1_{R_i}) \cdot x^{(i)} \in I$ since I is an ideal of the product ring, and so $\sum_{i=1}^n e_i(1_{R_i}) \cdot x^{(i)} \in I$ as well. However, $\sum_{i=1}^n e_i(1_{R_i}) \cdot x^{(i)} = (a_1, a_2, \ldots, a_n)$, so $(a_1, a_2, \ldots, a_n) \in I$. Therefore, $I \supseteq I_1 \times I_2 \times \cdots \times I_n$. For the converse inclusion $I \subseteq I_1 \times I_2 \times \cdots \times I_n$, let $(a_1, a_2, \ldots, a_n) \in I$. Then, $a_i = p_i(a_1, a_2, \ldots, a_n) \in I_i$ for all $1 \leq i \leq n$. Thus, $(a_1, a_2, \ldots, a_n) \in I_1 \times I_2 \times \cdots \times I_n$. Therefore, $I = I_1 \times I_2 \times \cdots \times I_n$. ■

We now wish to study the prime ideals of the product ring.

Proposition 1.51. *Let $R_1, R_2, \ldots, R_n$ be rings. Then $I \subseteq R_1 \times R_2 \times \cdots \times R_n$ is a prime ideal if and only if $p_i(I) = R_i$ for all $1 \leq i \leq n$ except for one index $1 \leq k \leq n$ such that $p_k(I)$ is a prime ideal of R_k.*

Proof. For the forward direction, let $I = I_1 \times \cdots I_n$ (by the previous proposition), where $I_i = p_i(I)$ is an ideal of R_i. Note that I is prime if and only if $(R_1 \times \cdots \times R_n)/I \cong R_1/I_1 \times \cdots R_n/I_n$ is an integral domain. This implies that R_i/I_i is an integral domain for one index $1 \leq k \leq n$ and $R_i/I_i \cong \{0\}$ for all $i \neq k$. It it were otherwise, we could find two indices $i \neq j$ such that R_i/I_i and R_j/I_j are both non-trivial, and $e_i(1_{R_i}) + I$ and $e_j(1_{R_j}) + I$ are two non-zero elements in $R_1/I_1 \times \cdots R_n/I_n$ whose product is zero, contradicting the fact that $R_1/I_1 \times \cdots R_n/I_n$ is an integral domain. Thus, $p_i(I) = I_i = R_i$ for all $i \neq k$, and $p_k(I) = I_k$ is a prime ideal of R_k.

Conversely, suppose that $p_i(I) = R_i$ for all $1 \leq i \leq n$ except for one index $1 \leq k \leq n$ such that $p_k(I)$ is a prime ideal of R_k. Then, $(R_1 \times \cdots \times R_n)/I \cong R_1/R_1 \times \cdots R_k/p_k(I) \times \cdots R_n/R_n \cong R_k/p_k(I)$ is an integral domain since $p_k(I)$ is a prime ideal of R_k. Thus, I is a prime ideal of the product ring. ■

1.6.1 Idempotents

Definition 1.52. Let R be a ring. An element $x \in R$ is called an *idempotent* if $x^2 = x$.

Example 1.53. Trivially, every ring has two idempotents, namely 0_R and 1_R. If R_1 and R_2 are rings, then $(1,1), (0,0), (1,0), (0,1)$ are idempotents of the product ring $R_1 \times R_2$.

Proposition 1.54. *Let R be a ring and $e \in R$ be an idempotent. Then $1 - e$ is also an idempotent, and $R \cong eR \times (1-e)R$.*

Proof. Note that $(1-e)e = e - e^2 = 0$, and $(1-e)^2 = (1-e)(1-e) = 1-e$. To show the isomorphism, let $\varphi : R \to eR \times (1-e)R$ be defined by $\varphi(r) = (er, (1-e)r)$ for all $r \in R$. It is easy to verify that φ is a ring homomorphism. If we define $\psi : eR \times (1-e)R \to R$ by $\psi(x,y) = x + y$ for all $x \in eR$ and $y \in (1-e)R$, then ψ is also a ring homomorphism. Moreover, $\psi \circ \varphi(r) = \psi(er, (1-e)r) = er + (1-e)r = r$ for all $r \in R$, and $\varphi \circ \psi(x,y) = \varphi(x+y) = (e(x+y), (1-e)(x+y)) = (x,y)$ for all $x \in eR$ and $y \in (1-e)R$. Thus, φ is an isomorphism, so $R \cong eR \times (1-e)R$. ■

February 9th.

An interesting consequence is the chinese remainder theorem, which is abstracted from the original number-theoretic version. For the below theorem, we define the product of two ideals I and J of a ring R as

$$IJ = \{a_1 b_1 + \cdots + a_n b_n \mid a_i \in I, b_i \in J, n \in \mathbb{N}\}. \tag{1.41}$$

One may verify that IJ is an ideal of R. Moreover, $IJ \subseteq I \cap J$ since $a_i b_i \in I$ and $a_i b_i \in J$ for all $1 \leq i \leq n$. However, it is not necessarily the case that $IJ = I \cap J$. For example, if $R = \mathbb{Z}$ and $I = (2)$ and $J = (4)$, then $IJ = (8)$ but $I \cap J = (4)$.

Theorem 1.55 (The *chinese remainder theorem*)**.** *Let R be a ring, and $I_1, I_2, \ldots, I_k$ be ideals of R which are pairwise co-maximal; that is, for all $1 \leq i \neq j \leq k$, we have $I_i + I_j = R$. Then $I_1 \cap I_2 \cap \cdots \cap I_k = I_1 I_2 \cdots I_k$. Moreover, the natural homomorphism $\varphi : R \to R/I_1 \times R/I_2 \times \cdots \times R/I_k$ defined by $\varphi(r) = (r + I_1, r + I_2, \ldots, r + I_k)$ is surjective with kernel $I_1 I_2 \cdots I_k$, so*

$$R/(I_1 I_2 \cdots I_k) \cong R/I_1 \times R/I_2 \times \cdots \times R/I_k. \tag{1.42}$$

We first determine how this abstract version of the chinese remainder theorem implies the original number-theoretic version. Let $n_1, n_2, \ldots, n_k$ be pairwise coprime positive integers, and let $m = n_1 n_2 \cdots n_k$. Then, the ideals $(n_1), (n_2), \ldots, (n_k)$ of $\mathbb{Z}$ are pairwise co-maximal since $(n_i) + (n_j) = (1)$ for all $1 \leq i \neq j \leq k$. Thus, by the chinese remainder theorem, we have

$$\mathbb{Z} \to \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}, \quad x \mapsto (x + n_1\mathbb{Z}, x + n_2\mathbb{Z}, \ldots, x + n_k\mathbb{Z}) \tag{1.43}$$

is surjective with kernel (m). That is, given any $(a_1 + n_1\mathbb{Z}, a_2 + n_2\mathbb{Z}, \ldots, a_k + n_k\mathbb{Z}) \in \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}$, there exists some integer x such that $x \equiv a_i \pmod{n_i}$ for all $1 \leq i \leq k$, and any two such integers are congruent modulo m. Moreover, we have the isomorphism

$$\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z} \times \cdots \times \mathbb{Z}/n_k\mathbb{Z}. \tag{1.44}$$

Proof. We first show for $k = 2$. Let I_1 and I_2 be two ideals of R such that $I_1 + I_2 = R$. Then, there exist $a \in I_1$ and $b \in I_2$ such that $a + b = 1$. We already know that $I_1I_2 \subseteq I_1 \cap I_2$, so we only need to show the converse inclusion. Let $x \in I_1 \cap I_2$. Then, $x = x(a + b) = xa + xb \in I_1I_2$ since $xa \in I_1I_2$ and $xb \in I_1I_2$. Therefore, $I_1 \cap I_2 = I_1I_2$. We now wish to show the natural map $R \to R/I_1 \times R/I_2$ defined by $r \mapsto (r + I_1, r + I_2)$ is surjective with kernel I_1I_2. It is easy to verify that the map is a ring homomorphism. For surjectivity, since any element of $R/I_1 \times R/I_2$ is of the form $(r_1 + I_1, r_2 + I_2)$ for some $r_1, r_2 \in R$, we only show that $(1 + I_1, 0 + I_2)$ and $(0 + I_1, 1 + I_2)$ are in the image of the map. Since $a \in I_1$, we have $a + I_1 = 0 + I_1$, so $\varphi(b) = (b + I_1, b + I_2) = (1 + I_1, 0 + I_2)$. Similarly, since $b \in I_2$, we have $b + I_2 = 0 + I_2$, so $\varphi(a) = (a + I_1, a + I_2) = (0 + I_1, 1 + I_2)$. Therefore, the map is surjective. For the kernel, if $r \in R$ is such that $\varphi(r) = (r + I_1, r + I_2) = (0 + I_1, 0 + I_2)$, then $r \in I_1 \cap I_2 = I_1I_2$. Conversely, if $r \in I_1I_2$, then $r \in I_1 \cap I_2$, so $\varphi(r) = (0 + I_1, 0 + I_2)$. Thus, the kernel of the map is I_1I_2. For arbitrary k, we claim that I_1 and $I_2 \cdots I_k$ are co-maximal ideals. Since I_1 and I_j are co-maximal for all $2 \leq j \leq k$, there exist $x_j \in I_1$ and $y_j \in I_j$ such that $x_j + y_j = 1$ for all $2 \leq j \leq k$. Then

$$(x_2 + y_2)(x_3 + y_3) \cdots (x_k + y_k) = 1 \implies \alpha + y_2y_3 \cdots y_k = 1 \tag{1.45}$$

where $\alpha \in I_1$ since $x_j \in I_1$ for all $2 \leq j \leq k$, and $y_2 \cdots y_k \in I_2 \cdots I_k$. Thus, I_1 and $I_2 \cdots I_k$ are co-maximal. Now induction can be applied on k to show $I_1 \cap I_2 \cap \cdots \cap I_k = I_1 \cap I_2 \cdots I_k = I_1I_2 \cdots I_k$. To show surjectivity of the natural map $R \to R/I_1 \times R/I_2 \times \cdots \times R/I_k$, we show $e_j := (0 + I_1, \ldots, 0 + I_{j-1}, 1 + I_j, 0 + I_{j+1}, \ldots, 0 + I_k)$ is in the image of the map for all $1 \leq j \leq k$. Since I_j and $I_1 \cdots I_{j-1}I_{j+1} \cdots I_k$ are co-maximal, there exist $x \in I_j$ and $y \in I_1 \cdots I_{j-1}I_{j+1} \cdots I_k$ such that $x + y = 1$. Then $\varphi(y) = (y + I_1, y + I_2, \ldots, y + I_k) = e_j$ since $y \in I_i$ for all $i \neq j$ and $y + I_j = 1 + I_j$. Therefore, the natural map is surjective with kernel $I_1I_2 \cdots I_k$, so we have the desired isomorphism $R/(I_1I_2 \cdots I_k) \cong R/I_1 \times R/I_2 \times \cdots \times R/I_k$. ■

1.7 Generalizing Properties of Integers

February 12th.

The ring of integers has several properties, which we axiomatize as properties for more general rings.

Definition 1.56. Let R be an integral domain. Let $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ be a function such that for all $a, b \in R \setminus \{0\}$, there exists $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $N(r) < N(b)$. Then, we say that R is a *euclidean domain* with respect to the euclidean function N.

Of course, the ring of integers $\mathbb{Z}$ is a euclidean domain with respect to the euclidean function $N : \mathbb{Z} \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(n) = |n|$ for all $n \in \mathbb{Z} \setminus \{0\}$. We know this to satisfy the condition that given any $a, b \in \mathbb{Z} \setminus \{0\}$, there exist $q, r \in \mathbb{Z}$ such that $a = bq + r$ and $0 \leq r < |b|$. If k denotes a field, then $k[x]$ is a euclidean domain with respect to the euclidean function $N : k[x] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(f) = \deg(f)$ for all $f \in k[x] \setminus \{0\}$. We know this to satisfy the condition that given any $f, g \in k[x] \setminus \{0\}$, there exist $q, r \in k[x]$ such that $f = gq + r$ and either $r = 0$ or $\deg(r) < \deg(g)$.

Example 1.57. A *valuation map* on a field k is a surjective map $v : k \to \mathbb{Z} \cup \{\infty\}$ such that $v(0) = \infty$, $v(ab) = v(a) + v(b)$ and $v(a + b) \geq \min\{v(a), v(b)\}$ for all $a, b \in k$. If v is a valuation map on k, then $R_v = \{x \in k : v(x) \geq 0\}$ is a valid ring with respect to the usual addition and multiplication on k termed the *valuation ring*. Moreover, R_v is a euclidean domain with respect to the euclidean function $N : R_v \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(x) = v(x)$ for all $x \in R_v \setminus \{0\}$. To show that N satisfies the condition in the definition of euclidean domain, let $a, b \in R_v \setminus \{0\}$. If $v(a) < v(b)$, then we can take $q = 0$ and $r = a$. If $v(a) \geq v(b)$, then we can take $q = a/b$ so that $v(a/b) = v(a) - v(b) \geq 0$, so $q \in R_v$, and $r = a - bq = 0$. Therefore, R_v is a euclidean domain with respect to the euclidean function N.

Definition 1.58. Let R be an integral domain. We say that R is a *principal ideal domain* if every ideal of R is principal; that is, for every ideal I of R, there exists some $a \in R$ such that $I = (a)$.

In other words, every ideal of R is generated by a single element. The ring of integers $\mathbb{Z}$ is a principal ideal domain since every ideal of $\mathbb{Z}$ is of the form (n) for some $n \in \mathbb{Z}$. More trivially, $\mathbb{Q}$ and $\mathbb{R}$ are principal ideal domains since the only ideals of $\mathbb{Q}$ and $\mathbb{R}$ are the zero ideal and the whole ring. Generally, we can show the following proposition.

Proposition 1.59. *Let R be a euclidean domain. Then R is a principal ideal domain.*

Proof. Let $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ be a euclidean function on R. Let I be an ideal of R. If $I = \{0\}$, then $I = (0)$ is principal. Else, let $x \in I$ be such that $N(x) = \min\{N(y) : y \in I \setminus \{0\}\}$. We claim that $I = (x)$. It is clear that $(x) \subseteq I$ since $x \in I$ and I is an ideal. For the converse inclusion, let $y \in I \setminus \{0\}$. Then, there exist $q, r \in R$ such that $y = xq + r$ and either $r = 0$ or $N(r) < N(x)$. If $r = 0$, then $y = xq \in (x)$. If $r \neq 0$, then $r = y - xq \in I$, contradicting the choice of x since $N(r) < N(x)$. Therefore, we must have $r = 0$, so $y \in (x)$. Thus, $I = (x)$ is principal. ■

Definition 1.60. Let R be a ring. A non-zero element $x \in R$ is called a *prime element* if the ideal (x) generated by x is a non-zero prime ideal of R. A non-zero element $x \in R$ is called an *irreducible element* if x is not a unit and whenever $x = ab$ for some $a, b \in R$, then either a or b is a unit.

That is, if $x \mid ab$, then $x \mid a$ or $x \mid b$ for all $a, b \in R$ (note that $x \mid a$ means there exists $c \in R$ such that $a = cx$) for a prime x. For example, in $\mathbb{Z}$, the prime elements and irreducible elements are exactly the prime numbers and their negatives. In $k[x]$ where k is a field, the prime elements and irreducible elements are exactly the irreducible polynomials.

Proposition 1.61. *Let R be an integral domain. Then every prime element of R is irreducible.*

Proof. Let $x = ab$ for $a, b \in R$. Then $x \mid ab$, so $x \mid a$ or $x \mid b$ since x is prime. If $x \mid a$, then there exists some $c \in R$ such that $a = cx$. Thus, $x = ab = cxb$, so $(1 - cb)x = 0$. Since R is an integral domain and $x \neq 0$, we must have $cb = 1$, so b is a unit. If $x \mid b$, then there exists some $d \in R$ such that $b = dx$. Thus, $x = ab = adx$, so $(1 - ad)x = 0$. Since R is an integral domain and $x \neq 0$, we must have $ad = 1$, so a is a unit. Therefore, either a or b is a unit, so x is irreducible. ■

Note that in a field, there are no prime elements. In a valuation ring, x is a prime element if and only if $v(x) = 1$.

Proposition 1.62. *Let R be a principal ideal domain. Then every irreducible element of R is prime.*

Proof. Consider $I = (x)$. Let $\mathfrak{m}$ be a maximal ideal of R containing I. Since R is a principal ideal domain, $\mathfrak{m} = (a)$ for some $a \in R$. Since $I \subseteq \mathfrak{m}$, we have $x \in (a)$, so there exists some $b \in R$ such that $x = ab$. Since x is irreducible, either a or b is a unit. If a is a unit, then $\mathfrak{m} = (a) = R$, contradicting the fact that $\mathfrak{m}$ is a maximal ideal. Thus, b is a unit and $a = b^{-1}x$. Thus $\mathfrak{m} = (a) = (b^{-1}x) \subseteq (x) = I$. Since $\mathfrak{m}$ is a maximal ideal containing I, we must have $\mathfrak{m} = I$. Therefore, I is a prime ideal, so x is a prime element. ■

Corollary 1.63. *In a principal ideal domain, every non-zero prime ideal is a maximal ideal.*

Definition 1.64. Let R be a ring, and $a, b \in R$. We say that $d \in R$ is a *greatest common divisor* of a and b, and we write $d = \gcd(a, b)$, if $d \mid a$ and $d \mid b$, and if there exists some $d' \in R$ such that $d' \mid a$ and $d' \mid b$, then $d' \mid d$.

Note that the gcd here may not be unique. For example, in $\mathbb{Z}$, the gcd of 6 and -9 is 3 and -3.

Proposition 1.65. *Let R be a ring, and $a, b \in R$ such that (a, b) is a principal ideal. Then the* gcd *of a and b exists, and if $(d) = (a, b)$, then d is a* gcd *of a and b. Moreover, $d = ax + by$ for some $x, y \in R$.*

Proof. If $(a,b) = (d)$, then $d \mid a$ and $d \mid b$. If $d' \in R$ is such that $d' \mid a$ and $d' \mid b$, then $(a,b) \subseteq (d')$, so $(d) \subseteq (d')$, so $d' \mid d$. Thus, d is a gcd of a and b. Since $(d) = (a,b)$, there exist $x, y \in R$ such that $d = ax + by$. ■

Of course, the notion of gcd exists for $\mathbb{Z}$. As another example, look at this ring $\mathbb{Q}[x,y]$. By inspection, 1 is a gcd of x and y, since $1 \mid x$ and $1 \mid y$, and if $d' \in \mathbb{Q}[x,y]$ is such that $d' \mid x$ and $d' \mid y$, then $d' \in \mathbb{Q}$, so $d' \mid 1$. In fact, any $a \in \mathbb{Q}^\times$ is a gcd of x and y.

February 13th.

Example 1.66. Consider the rings $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-3}]$. The ring $\mathbb{Z}[i]$ is a euclidean domain with respect to the euclidean function $N : \mathbb{Z}[i] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $N(a+bi) = a^2 + b^2$ for all $a + bi \in \mathbb{Z}[i] \setminus \{0\}$. Indeed, given any $\alpha = a + ib, \beta = c + id \in \mathbb{Z}[i]$, for $a, b, c, d \in \mathbb{Z}$, we wish to find $\gamma, r \in \mathbb{Z}[i]$ such that $\alpha = \beta\gamma + r$ and either $r = 0$ or $N(r) < N(\beta)$. Consider

$$\frac{\alpha}{\beta} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = u + iv \tag{1.46}$$

where $u, v \in \mathbb{Q}$. So there exist $p, q \in \mathbb{Z}$ such that $|p - u| \leq 1/2$ and $|q - v| \leq 1/2$. Thus,

$$\alpha = \beta(u+iv) = \beta(p+iq) + \beta((u-p) + i(v-q)) = \beta(p+iq) + r \tag{1.47}$$

where $N(r) = N(\beta)N((u-p)+i(v-q)) \leq N(\beta)\frac{1}{2} < N(\beta)$. Thus, $\gamma = p+iq$ and $r = \beta((u-p)+i(v-q))$ satisfy the required condition. Therefore, $\mathbb{Z}[i]$ is a euclidean domain, so $\mathbb{Z}[i]$ is a principal ideal domain. Thus, $\mathbb{Z}[i]$ is a principal ideal domain, so every irreducible element of $\mathbb{Z}[i]$ is prime. In particular, $1+i$ is an irreducible element of $\mathbb{Z}[i]$, so $1+i$ is prime.
On the other hand, the ring $\mathbb{Z}[\sqrt{-3}]$ is not a principal ideal domain; consider $I = (2, 1+\sqrt{-3})$. We show that $I \cap \mathbb{Z} = (2)$. Since $2 \in I$, we have $(2) \subseteq I \cap \mathbb{Z}$. For the converse inclusion, let $x \in I \cap \mathbb{Z}$. Then, there exist $r, s \in \mathbb{Z}[\sqrt{-3}]$ such that $x = r(1+\sqrt{-3}) + s2$. Since x is also an integer, we can write $x = \alpha(1+\sqrt{-3}) + 2\beta$ where $\alpha \in \mathbb{Z}[-\sqrt{3}]$ and $\beta \in \mathbb{Z}$. But this means that $\alpha(1+\sqrt{-3})$ must be an integer, so taking norms gives $(x-2\beta)^2 = \alpha\bar{\alpha}4$, so $4 \mid (x-2\beta)^2$, so $2 \mid x - 2\beta$, so $2 \mid x$. Thus, $x \in (2)$, so $I \cap \mathbb{Z} = (2)$. Hence, $I \subsetneq \mathbb{Z}[\sqrt{-3}]$. If I were principal, then $I = (a + b\sqrt{-3})$ for some $a, b \in \mathbb{Z}$. If $b = 0$, then $I = (a)$ implies $I = 2\mathbb{Z}[\sqrt{-3}]$ which makes the contradiction $1 + \sqrt{-3} \notin 2\mathbb{Z}[\sqrt{-3}]$; b has to be non-zero. We must have, as $2 \in I$, $2 = (a + b\sqrt{-3})(c + d\sqrt{-3})$ for some $c, d \in \mathbb{Z}$. Taking norms gives $4 = (a^2 + 3b^2)(c^2 + 3d^2)$. Since $b \neq 0$, $b = \pm 1$ and $a = \pm 1$ must be the only possibilities. This again must mean that $c^2 + 3d^2 = 1$ which implies $c = \pm 1$ and $d = 0$. But this gives $2 = \pm(1 \pm \sqrt{-3})$, which is nonsense. Hence I cannot be principal.

A more general result states that if $D \equiv 1 \pmod 4$ and D is square-free, then $\mathbb{Z}[\sqrt{D}]$ cannot be a principal ideal domain.

Proposition 1.67. *Let R be a ring such that $R[x]$ is a principal ideal domain. Then R is a field.*

Proof. Since $R \subseteq R[x]$ is a subring, R is an integral domain. Let $\varphi : R[x] \to R$ be the evaluation homomorphism defined by $\varphi(f) = f(0)$ for all $f \in R[x]$. This map is surjective. Moreover, $\ker \varphi = (x) \subseteq R[x]$. So $R[x]/(x) \cong R$, implying that $R[x]/(x)$ is an integral domain and (x) is a prime ideal. We had shown that any non-zero prime ideal in a principal ideal domain is maximal, so (x) is maximal. Thus, $R[x]/(x) \cong R$ is a field, so R is a field. ■

Definition 1.68. For a ring R, the map $N : R \to \mathbb{Z}_{\geq 0}$ is called a *Dedekind-Hasse* norm if $N(0) = 0$, and for all $a, b \in R \setminus \{0\}$, either $b \mid a$ or there exists $0 \neq r \in (a,b)$ such that $N(r) < N(b)$.

Proposition 1.69. *If R is an integral domain that admits a Dedekind-Hasse norm, then R is a principal ideal domain. The converse also holds.*

Proof. Let $I \subseteq R$ be an ideal. Let $x \in I \setminus \{0\}$ be such that $N(x)$ is the least among $\{N(y) \mid y \in I, y \neq 0\}$. $(x) \subseteq I$ is clear. Let $y \in I \setminus \{0\}$. So either $x \mid y$ or there exists $0 \neq r \in (x, y)$ such that $N(r) < N(x)$. This, however, is a contradiction to the choice of x. Hence $x \mid y$ and $y \in (x)$. Thus, $I = (x)$ is principal. ■

Example 1.70. The ring $\mathbb{Z}[(1+\sqrt{-19})/2]$ is a principal ideal domain but not a euclidean domain. Note that $N : \mathbb{Z}[\omega = (1+\sqrt{-19})/2] \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ defined by $a+b\omega \mapsto a^2+ab+5b^2$ is a Dedekind-Hasse norm.

Proposition 1.71. *Let R be a euclidean domain but not a field. Then it contains a* 'universal side divisor'*; that is, there exists a non-zero non-unit $u \in R$ such that for all non-zero $x \in R$, there exists $q \in R$ satisfying that $r = x - qu$ is either 0 or a unit*

Proof. Let $u \in \mathbb{R} \setminus \{0\}$ be such that $N(u)$ is least among all non-zero non-unit elements of R. We claim that u satisfies this property. Let $x \in R \setminus \{0\}$. Then there exist $q, r \in R$ such that $x = qu + r$ satisfying either $r = 0$ or $N(r) < N(u)$. Since u is a non-unit, $N(u) > 0$, so $N(r) < N(u)$ implies that r is a unit. Thus, $r = x - qu$ is either 0 or a unit, so the claim holds. ■

One may check that $\mathbb{Z}[(1+\sqrt{-19})/2]$ has no universal side division, and ± 1 are the only units in the ring.

Definition 1.72. Let R be an integral domain. We say that R is a *unique factorization domain* if for all non-zero non-unit $x \in R$, x can be written as $x = p_1 \cdots p_n$ for some $n \geq 1$ and p_i irreducibles, and if $x = p_1 \cdots p_n = q_1 \cdots q_m$ for some $m \geq 1$ and irreducibles q_j, then $n = m$ and, after reordering, $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$.

For example, both $\mathbb{Z}$ and $k[x]$ where k is a field are UFDs. As a small definition, $x, y \in R$ are termed *associates* if $x = uy$ for some unit $u \in R$. Again, in $\mathbb{Z}$, $-n$ and n are associates, and in $\mathbb{Q}[x]$, f and cf for some $c \in \mathbb{Q}^\times$ are associates. Lastly, in $\mathbb{Z}[i]$, $n, -n, ni, -ni$ are all associates for any $n \in \mathbb{Z}$. Thus the above definition is saying that the factorization of x into irreducibles is unique up to reordering and up to associates. For non-examples, $\mathbb{Z}[\sqrt{5}]$ is an integral domain but not a unique factorization domain since $-4 = -2 \cdot 2 = (1+\sqrt{5})(1-\sqrt{5})$ are two distinct factorizations of -4 into irreducibles, and the irreducibles $2, 1+\sqrt{5}, 1-\sqrt{5}$ are not associates of each other.

February 16th.

Example 1.73. As a more concrete example of a non-UFD, consider the quotient ring $\bar{R} = \mathbb{Q}[x, y, z, w]/(xy - zw)$. Suppose $\bar{x} = \bar{f}\bar{g}$ for some $\bar{f}, \bar{g} \in \bar{R}$. Then $x - fg \in (xy - zw)$ unless $x - fg = 0$. Since $xy - zw$ is irreducible in $\mathbb{Q}[x, y, z, w]$, we must have $x \mid f$ or $x \mid g$. If $x \mid f$, then $\bar{f}$ is a unit in $\bar{R}$, so $\bar{x}$ is irreducible. Similarly, if $x \mid g$, then $\bar{g}$ is a unit in $\bar{R}$, so $\bar{x}$ is irreducible. Thus, $\bar{x}$ is irreducible. By symmetry, $\bar{y}, \bar{z}, \bar{w}$ are also irreducible. However, we have the two distinct factorizations $\bar{x}\bar{y} = \bar{z}\bar{w}$ into irreducibles, and the irreducibles $\bar{x}, \bar{y}, \bar{z}, \bar{w}$ are not associates of each other since they are all of degree one. Therefore, $\bar{R}$ is not a unique factorization domain.

Theorem 1.74. *Let R be a principal ideal domain. Then R is a unique factorization domain.*

Proof. Let $x \in R$ be a non-zero non-unit. We show that x can be written as a product of irreducibles. Let $\mathfrak{m}$ be a maximal ideal of R containing (x). Since R is a principal ideal domain, $\mathfrak{m} = (p_1)$ for some $p_1 \in R$. Thus, p_1 is a prime element, so p_1 is irreducible. If $x = p_1 x_1$ for some $x_1 \in R$, then we can repeat the process for x_1. If x_1 is a unit, then we are done. Else, we can repeat the process for x_1 to get a prime element p_2 such that $x_1 = p_2 x_2$ for some $x_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots$. Let $I = \bigcup_{n \geq 0} (x_n)$. Then I is an ideal of R. Since R is a principal ideal domain, $I = (y)$ for some $y \in R$. Thus, $y \in I$ implies that $y \in (x_r)$ for some $r \geq 0$, so $y = ux_r$ for some $u \in R$. Also, $x_r = p_{r+1} x_{r+1}$ for some $p_{r+1} \in R$ and $x_{r+1} \in R$, so $y = up_{r+1} x_{r+1}$. Also, since $x_{r+1} = vy$ for some $v \in R$, we have $y = up_{r+1} vy$, so $(1 - up_{r+1} v)y = 0$. Since R is an integral domain and $y \neq 0$, we must have $up_{r+1} v = 1$, so p_{r+1} is a unit, contradicting the fact that p_{r+1} is a prime element. Therefore, the

process must terminate after finitely many steps, so x can be written as a product of irreducibles (note that the last terminating unit may be absorbed into the previous prime element to make it irreducible).

To show uniquness upto reordering, and associates, let $x = p_1 \cdots p_n = q_1 \cdots q_m$ for some $m \geq 1$ and irreducibles q_j. Since p_1 is irreducible and R is a principal ideal domain, p_1 is prime, so $p_1 \mid q_1 \cdots q_m$ implies that $p_1 \mid q_{i_1}$ for some $1 \leq i_1 \leq m$. Since q_{i_1} is irreducible, p_1 and q_{i_1} are associates, so $p_1 = u_1 q_{i_1}$ for some unit u_1. We can reorder the q_j's so that $i_1 = 1$, giving us $p_1 = u_1 q_1$. Cancellation gives us $p_2 \cdots p_n = u_1 q_2 \cdots q_m$; we can repeat the same process to get $p_2 = u_2 q_2$ for some unit u_2, and so on. After repeating this process, we get $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$. By symmetry argument, we also get $q_j = v_j p_j$ for some unit v_j for all $1 \leq j \leq m$. Thus, $n = m$ and $p_i = u_i q_i$ for some unit u_i for all $1 \leq i \leq n$, so the factorization of x into irreducibles is unique up to reordering and up to associates. ■

Proposition 1.75. *Let R be a unique factorization domain. Then an element is irreducible if and only if it is prime.*

Proof. We already know that every prime element is irreducible in any integral domain. For the converse, let x be an irreducible element of R. Let $x \mid ab$ for some $a, b \in R$. There then exists some $y \in R$ such that $ab = xy$. Let $a = p_1 \cdots p_n$ and $b = q_1 \cdots q_m$, and $y = r_1 \cdots r_k$ be the factorizations of a, b, y into irreducibles. Then $p_1 \cdots p_n q_1 \cdots q_m = r_1 \cdots r_k x$. Since x is irreducible, x must be an associate of some p_i or some q_j, so $x \mid a$ or $x \mid b$. Thus, x is prime. ■

Theorem 1.76. *Let R be a principal ideal domain. Then R has a Dedekind-Hasse norm.*

Proof. Let $x \in R \setminus \{0\}$. Define $N : R \setminus \{0\} \to \mathbb{Z}_{\geq 0}$ as follows: if x is a unit, $N(x) = 1$. Otherwise, write $x = p_1 \cdots p_n$ for some $n \geq 1$ and irreducibles p_i, and define $N(x) = 2^n$. Verify that N is a Dedekind-Hasse norm. ■

February 19th.

Proposition 1.77. *Let R be a unique factorization domain. If $a, b \in R$ are non-zero, then $\gcd(a, b)$ exists. If a or b is a unit, then $\gcd(a, b)$ is a unit. Moreover, if a and b are non-units, then there exist irreducibles $p_1, \ldots, p_r$ such that $a = p_1^{e_1} \cdots p_r^{e_r}$ and $b = p_1^{f_1} \cdots p_r^{f_r}$ for some $e_i, f_i \geq 0$ for all $1 \leq i \leq r$, and $\gcd(a, b) = p_1^{\min\{e_1, f_1\}} \cdots p_r^{\min\{e_r, f_r\}}$.*

Proof. If a or b is a unit, then $\gcd(a, b)$ is a unit since any unit divides any element. So we may assume that a and b are non-units. Since R is a unique factorization domain, there exist irreducibles $p_1, \ldots, p_r$ such that $a = p_1^{e_1} \cdots p_r^{e_r}$ and $b = p_1^{f_1} \cdots p_r^{f_r}$ for some $e_i, f_i \geq 0$ for all $1 \leq i \leq r$. Let $d = p_1^{\min\{e_1, f_1\}} \cdots p_r^{\min\{e_r, f_r\}}$. Then $d \mid a$ and $d \mid b$. If $d' \in R$ is such that $d' \mid a$ and $d' \mid b$, then there exist some irreducibles $q_1, \ldots, q_s$ such that $d' = q_1^{g_1} \cdots q_s^{g_s}$ for some $g_i \geq 0$ for all $1 \leq i \leq s$. Since $d' \mid a$, we have that each irreducible factor of d' must be an associate of some irreducible factor of a, so after reordering, we can write $d' = p_1^{g_1'} \cdots p_r^{g_r'}$ where each $g_i' \geq 0$. Since $d' \mid b$, we have that each irreducible factor of d' must be an associate of some irreducible factor of b, so after reordering, we can write $d' = p_1^{g_1''} \cdots p_r^{g_r''}$ where each $g_i'' \geq 0$. Thus, after reordering, we can write $d' = p_1^{g_1'''} \cdots p_r^{g_r'''}$ where $g_i''' = \min\{g_i', g_i''\} \leq \min\{e_i, f_i\}$ for all $1 \leq i \leq r$. Thus, $d' \mid d$. Therefore, d is a gcd of a and b. ■

Proposition 1.78. *Let R be an integral domain. Then R is a unique factorization domain if and only if every irreducible element of R is prime and any increasing chain of principal ideals in R stabilizes.*

Proof. For the foward implication, if have already shown that every irreducible element of R is prime. Let $(x_1) \subsetneq (x_2) \subsetneq (x_3) \subsetneq \cdots$ be a strictly increasing chain of principal ideals in R. If x_1 is a unit, then $(x_1) = R$, contradicting the fact that $(x_1) \subsetneq (x_2)$. If $x_1 = 0$, then $x_2 \neq 0$ so we may assume that x_1 is a non-zero non-unit. Since R is a unique factorization domain, we can uniquely factor x_1 as $x_1 = p_1 \cdots p_n$ for some $n \geq 1$ and irreducibles p_i. Since $x_1 \in (x_2)$, $x_1 = x_2 y$ for some $y \in R$. Also $(x_1) \subsetneq (x_2)$ implies that y is not a unit. Let $x_2 = q_1 \cdots q_m$ and $y = r_1 \cdots r_k$ be the factorizations of x_2 and y into irreducibles. Then by the uniquness of factorization, $n = m + k$ and since y is not a unit, $k \geq 1$, so $m < n$. Thus, x_2 has strictly less irreducible factors than x_1. We can repeat the same process for x_2 to get that x_3 has

strictly less irreducible factors than x_2, and so on. Since the number of irreducible factors is a non-negative integer, this process must terminate after finitely many steps giving a x_n that is a unit, contradicting the fact that $(x_n) \subsetneq (x_{n+1})$. Therefore, any increasing chain of principal ideals in R stabilizes.

For the converse implication, let $x \in R$ be a non-zero non-unit. The result is trivial if x is irreducible, so we may assume that x is not an irreducible. We claim that there exists an irreducible $p_1 \in R$ such that $x = p_1 y$ for some $y \in R$. Since x is not irreducible, there exist $x_1, y_1 \in R$ such that $x = x_1 y_1$ and neither x_1 nor y_1 is a unit. If x_1 is irreducible, then we can take $p_1 = x_1$ and $y = y_1$. Else, we can repeat the same process for x_1 to get $x = x_2 y_2 y_1$ for some non-units $x_2, y_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (x_1) \subsetneq (x_2) \subsetneq \cdots$. This contradicts the fact that any increasing chain of principal ideals in R stabilizes. Therefore, the process must terminate after finitely many steps, so there exists an irreducible $p_1 \in R$ such that $x = p_1 y$ for some $y \in R$. If y is a unit, then we are done. Else, we can repeat the same process for y to get $x = p_1 p_2 y_2$ for some irreducible p_2 and some $y_2 \in R$. We can keep repeating this process; we claim that it must terminate after finitely many steps. If not, then we have an infinite sequence of ideals $(x) \subsetneq (p_1) \subsetneq (p_1 p_2) \subsetneq \cdots$. This contradicts the fact that any increasing chain of principal ideals in R stabilizes. Therefore, the process must terminate after finitely many steps, so x can be written as a product of irreducibles. The uniquenss follows from the same proof as that of the theorem that a principal ideal domain is a unique factorization domain. ■

1.8 Localization

March 2nd.

Localization of rings is a process that allows us to 'invert' a subset of elements in a ring. We will see that the localization of $\mathbb{Z}$ leads to $\mathbb{Q}$.

Definition 1.79. Let R be an integral domain. Then $K := \{(a,b) \equiv \frac{a}{b} \mid a, b \in R,\ b \neq 0\} = R \times (R \setminus \{0\})/\sim$ where $(a,b) \sim (c,d)$ if and only if $ad = bc$ is a field, called the *field of fractions* of R; the operations in this field are defined as follows: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ and $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$.

We may denote the field of fractions of R by $QF(R)$ or $\mathrm{frac}(R)$. Note that the field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$; the definition is simply abstracted from the construction of $\mathbb{Q}$ from $\mathbb{Z}$. Note that we have a natural embedding $R \hookrightarrow K$ given by $a \mapsto \frac{a}{1}$ for all $a \in R$. We will identify R with its image in K under this embedding, so we will view R as a subring of K. If R itself is a field, then R is its own field of fractions. We wish to generalize this idea; that is, we want to 'invert' only a particular subset of elements in R instead of all non-zero elements.

Definition 1.80. Let R be a ring. A subset $S \subseteq R$ is called a *multiplicative subset* if $1 \in S$ and if $s, t \in S$, then $st \in S$.

For example, if R is a ring then any subring S works as a multiplicative subset. If $R = \mathbb{Z}$, we can take $S = \mathbb{N}$. We can also take S to be $R^\times$, the group of units. If R is an integral domain and $S = R \setminus \{0\}$, then S is a multiplicative subset. For any R, we can also take $S = I \setminus \{0\} \cup \{1\}$ for any ideal I of R. In particular, we are interested in this example: if R is a ring and P is a prime ideal of R, then $S = R \setminus P$ is a multiplicative subset. To show this, let $x, y \in R \setminus P$. If $xy \in P$, then since P is prime, either $x \in P$ or $y \in P$, contradicting the choice of x, y. Thus, $xy \in R \setminus P$. Moreover, P is a proper ideal, so $1 \in R \setminus P$. Thus, $R \setminus P$ is a multiplicative subset. Similarly, $S = R \setminus \bigcup_{i=1}^n P_i$ is a multiplicative subset for any prime ideals $P_1, \ldots, P_n$ of R.

Definition 1.81. Let R be a ring and $S \subseteq R$ be a multiplicative subset. Then on $S \times R$, we define an (equivalence) relation $\sim$ by $(s_1, r_1) \sim (s_2, r_2)$ if and only if there exists some $t \in S$ such that $t(s_1 r_2 - s_2 r_1) = 0$. We can verify that $\sim$ is an equivalence relation. Let $S^{-1}R = S \times R/\sim$. We define $\oplus$ and $\odot$ on $S^{-1}R$ as follows: $\frac{r_1}{s_1} \oplus \frac{r_2}{s_2} = \frac{s_2 r_1 + s_1 r_2}{s_1 s_2}$ and $\frac{r_1}{s_1} \odot \frac{r_2}{s_2} = \frac{r_1 r_2}{s_1 s_2}$. These operations are well-defined, and $S^{-1}R$ is a ring with respect to these operations, called the *localization* of R at S. Finally, we have a natural homomorphism $\varphi : R \to S^{-1}R$ defined by $\varphi(r) = \frac{r}{1}$ for all $r \in R$.

March 5th.

We will simply use $+$ and $\cdot$ for $\oplus$ and $\odot$ respectively. We verify some of the claims made in this definition.

Proof. That $\sim$ is reflexive and symmetric is clear. To show that $\sim$ is transitive, let $(s_1, r_1) \sim (s_2, r_2)$ and $(s_2, r_2) \sim (s_3, r_3)$. Then there exist some $t, t' \in S$ such that $t(s_1r_2 - s_2r_1) = 0$ and $t'(s_2r_3 - s_3r_2) = 0$. Then, multiplying the first equation by s_3t' and the second equation by s_1t, we get $ts_3t'(s_1r_2 - s_2r_1) = 0$ and $s_1tt'(s_2r_3 - s_3r_2) = 0$. Adding these two equations gives us $ts_3t'(s_1r_2 - s_2r_1) + s_1tt'(s_2r_3 - s_3r_2) = 0$, so $tt'(s_1s_3r_2 - s_1s_2r_3 + s_1s_2r_3 - s_1s_3r_2) = 0$, so $tt'(s_1s_3r_2 - s_1s_3r_2) = 0$, so $tt' \cdot 0 = 0$. Thus, $(s_1, r_1) \sim (s_3, r_3)$, so $\sim$ is transitive. $\sim$ is an equivalence relation.

To show that $\odot$ is well-defined, let $\frac{r_1}{s_1} = \frac{r_1'}{s_1'}$ and $\frac{r_2}{s_2} = \frac{r_2'}{s_2'}$. Then there exist some $u, v \in S$ such that $u(s_1'r_1 - s_1r_1') = 0$ and $v(s_2'r_2 - s_2r_2') = 0$. Then $uv(s_1's_2'r_1r_2 - s_1s_2r_1r_2') = uv(s_1'r_1 \cdot s_2'r_2 - s_1r_1' \cdot s_2r_2') = uv(s_1'r_1 \cdot s_2'r_2 - s_1'r_1 \cdot s_2r_2' + s_1'r_1 \cdot s_2r_2' - s_1r_1' \cdot s_2r_2') = uv(s_1'r_1 \cdot (s_2'r_2' - s_2'r_2') + (s_1'r_1' - s_1'r_1') \cdot s_2'r_2') = uv(0+0) = 0$. Thus, $\frac{r_1r_2}{s_1s_2} = \frac{r_1'r_2'}{s_1's_2'}$, so $\odot$ is well-defined. Similarly, we can show that $\oplus$ is well-defined.

Showing that $(S^{-1}R, \oplus)$ is an abelian group is straightforward, with the identity element being $\frac{0}{1}$ and the inverse of $\frac{r}{s}$ being $\frac{-r}{s}$. The distributivity of $\odot$ over $\oplus$, and its associativity and commutativity, are also straightforward to verify, with $\frac{1}{1}$ being the multiplicative identity. Thus, $S^{-1}R$ is a ring with respect to these operations. Finally, φ may be verified to be a homomorphism by checking that $\varphi(r_1 + r_2) = \varphi(r_1) \oplus \varphi(r_2)$ and $\varphi(r_1r_2) = \varphi(r_1) \odot \varphi(r_2)$ for all $r_1, r_2 \in R$. ■

Example 1.82. The most common example, of course, is to take $\mathbb{R} = \mathbb{Z}$ and $S = \mathbb{Z} \setminus \{0\}$. Then $S^{-1}R$ is the field of fractions of $\mathbb{Z}$, which is $\mathbb{Q}$. More generally, if R is an integral domain and $S = R \setminus \{0\}$, then $S^{-1}R$ is the field of fractions of R. In such a case, the natural homomorphism $\varphi : R \to S^{-1}R$ is injective, so we can identify R with its image in $S^{-1}R$, and view R as a subring of $S^{-1}R$. Thus, we can view the field of fractions of an integral domain as an extension of the integral domain.

Proposition 1.83. *Let $\varphi : R \to S^{-1}R$ be the natural homomorphism. Then $\ker \varphi = \{r \in R \mid sr = 0$ for some $s \in S\}$. In particular, if S consists only of non-zero divisors, then φ is injective.*

Proof. The second statement follows from the first statement since $\ker \varphi = \{0\}$ if S consists only of non-zero divisors. To show the first statement, let $r \in R$. If $r \in \ker \varphi$, then $\varphi(r) = \frac{r}{1} = 0$ in $S^{-1}R$, so there exists some $s \in S$ such that $s(r \cdot 1 - 0 \cdot 1) = sr = 0$. Conversely, if $sr = 0$ for some $s \in S$, then $\varphi(r) = \frac{r}{1} = \frac{0}{s} = 0$ in $S^{-1}R$, so $r \in \ker \varphi$. Thus, $\ker \varphi = \{r \in R \mid sr = 0 \text{ for some } s \in S\}$. ■

From here, we verify that $S^{-1}R$ indeed gives the field of fractions for an integral domain R by verifying that it is a field; indeed; let $\frac{r}{s} \in S^{-1}R$ such that $\frac{r}{s} \neq \frac{0}{1}$; this occurs iff $t(r - 0) \neq 0$ for all $t \in S$ which occurs iff $r \neq 0$, since R is an integral domain. This means that $r \in S$ so $\frac{s}{r} \in S^{-1}R$ showing that $\frac{r}{s}$ has a multiplicative inverse.

An important example of a multiplicative subset of a ring R is to choose $S = \{1, x, x^2, x^3, \ldots\}$ for some $x \in R$. In such a case, we denote $S^{-1}R = R[\frac{1}{x}]$. In particular, if we choose $R = \mathbb{Z}$ and $S = \{1, 2, 4, 8, \ldots\}$, then we get $S^{-1}R = \mathbb{Z}[\frac{1}{2}]$. If R is an integral domain, we can write $R[\frac{1}{x}]$ as $\{\frac{a}{x^k} \mid a \in R,\ k \geq 0\}$. Some choices of S can lead to rather ugly constructions; suppose $0 \in S$ for the ring R. Then $\frac{r_1}{s_1}, \frac{r_2}{s_2} \in S^{-1}R$ are always equal; $0 \cdot (s_1r_2 - s_2r_1) = 0$ since $0 \in S$, making the construction a trivial ring.

Proposition 1.84. *Let R be a ring, S a multiplicative subset, and $\varphi : R \to S^{-1}R$ be the localization homomorphism. Then for $s \in S$, $\varphi(s)$ is a unit in $S^{-1}R$.*

Proof. s is mapped to $\frac{s}{1}$. Since $s \in S$, $\frac{1}{s} \in S^{-1}R$, which is the multiplicative inverse of $\frac{s}{1}$; $\varphi(s)$ is a unit. ■

Let R be an integral domain, and let $S_x = \{1, x, x^2, \ldots\}$ and $S_y = \{1, y, y^2, \ldots\}$ for some $x, y \in R$. Also let $S_{xy} = \{1, xy, x^2y^2, \ldots\}$. Then, we have $S_x^{-1}R \subseteq S_{xy}^{-1}R$.

March 6th.

Recall that we define the quotient map as follows: given any ring homomorphism $\varphi : R \to A$ such that $I \subseteq \ker \varphi$ for I an ideal of R, then there exists a unique map $\bar{\varphi} : R/I \to A$ such that $\bar{\varphi} \circ q = \varphi$, via $\bar{\varphi}(r + I) = \varphi(r)$. One can also define the quotient ring as follows.

Definition 1.85. A quotient of a ring R by an ideal I is a ring homomorphism $q : R \to B$ such that for any ring homomorphism $\varphi : R \to A$ with $I \subseteq \ker \varphi$, there exists a unique ring homomorphism $\bar{\varphi} : B \to A$ satisfying $\bar{\varphi} \circ q = \varphi$.

Theorem 1.86 (The *universal property of localization*)**.** *Let R be a ring and S a multiplicative subset. Also let $f : R \to A$ be a ring homomorphism such that $f(s)$ is a unit in A for all $s \in S$. Then there exists a unique homomorphism $\bar{f} : S^{-1}R \to A$ such that $\bar{f} \circ \varphi = f$, where $\varphi : R \to S^{-1}R$ is the localization map.*

Proof. We define $\bar{f} : S^{-1}R \to A$ as $\bar{f}(\frac{r}{s}) = f(s)^{-1}f(r)$. This $\bar{f}$ is well-defined; to show this let $\frac{r}{s} = \frac{r'}{s'}$. That is, there exists $u \in S$ such that $u(s'r - sr') = 0$. Thus,

$$f(u)(f(s')f(r) - f(s)f(r')) = 0 \implies f(s')f(r) = f(s)f(r') \implies \bar{f}(\frac{r}{s}) = \bar{f}(\frac{r'}{s'}) \tag{1.48}$$

since $u, s, s' \in S$ are all mapped to units in A. To show $\bar{f}$ is a ring homomorphism, we first have

$$\bar{f}(\frac{r_1}{s_1} + \frac{r_2}{s_2}) = \bar{f}(s_1s_2)^{-1}f(s_2r_1 + s_1r_2) = f(s_1)^{-1}f(r_1) + f(s_2)^{-1}f(r_2) = \bar{f}(\frac{r_1}{s_1}) + \bar{f}(\frac{r_2}{s_2}). \tag{1.49}$$

Similarly, one can show multiplication. For uniqueness, let $g : S^{-1}R \to A$ be a ring homomorphism such that $g \circ \varphi = f$. Then $g(\frac{r}{s}) = g(\frac{r}{1})g(\frac{1}{s}) = g \circ \varphi(r)(g \circ \varphi(s))^{-1} = f(r)f(s)^{-1} = \bar{f}(\frac{r}{s})$ showing $g = \bar{f}$. ■

We look at some applications of localization.

Proposition 1.87. *Let R be an integral domain and K be a field. Moreover, let $i : R \to K$ be an injective ring homomorphism. Then i extends to an injective ring homomorphism $\tilde{\iota} : \mathrm{frac}(R) \to K$. In other words, if $R \subseteq K$, then $\mathrm{frac}(R) \subseteq K$.*

Here, recall that $\mathrm{frac}(R)$ is the field of fractions of an integral domain R; its the localization of R with $R \setminus \{0\}$ chosen as the multiplicative subset.

Proof. Note that $i(s)$ is a unit in K for all $s \neq 0$. By the universal property, there exists a unique ring homomorphism $\tilde{\iota} : \mathrm{frac}(R) \to K$ such that $\tilde{\iota} \circ \varphi = i$, with φ being the localization map. Thus, $\tilde{\iota}(\frac{r}{1}) = i(r)$ for all $r \in R$. Moreover, $\tilde{\iota} : \mathrm{frac}(R) \to K$ is also injective; we have

$$\tilde{\iota}(\frac{r}{s}) = 0 \implies \tilde{\iota}(\frac{r}{1}) = i(r) = 0 \implies r = 0 \implies \frac{r}{s} = \frac{0}{1} \tag{1.50}$$

since i is injective. ■

Proposition 1.88. *Let R be a ring and $a \in R$. Then,*

$$R[X]/(aX - 1) \cong R[\frac{1}{a}] \tag{1.51}$$

via the map $r \mapsto \frac{r}{1}$ for all $r \in R$ and $X \mapsto \frac{1}{a}$.

Proof. Define the map $\alpha : R[X] \to S^{-1}R = R[\frac{1}{a}]$ via $f(X) \mapsto f(\frac{1}{a})$. Let $I = (aX - 1)$. Note that $\alpha(aX - 1) = \frac{a}{1} \cdot \frac{1}{a} - \frac{1}{1} = 0$, so $(aX - 1) \subseteq \ker \alpha$. Thus, we have a ring homomorphism $\bar{\alpha} : R[X]/I \to S^{-1}R$ such that $\bar{\alpha} \circ q = \alpha$, q being the quotient map; $\bar{\alpha}(f(X) + I) = f(\frac{1}{a})$. We now give a map in the other direction; let $f : R \to R[X]/I$ be the map defined by $f(r) = r + I$. Note that $S = \{1, a, a^2, \ldots\}$; each a^n is mapped to the unit $a^n + I$ with inverse $X^n + I$. Thus, by the universal property of localization, there exists a unique ring homomorphism $\bar{f} : S^{-1}R \to R[X]/(aX - 1)$ such that $\bar{f} \circ \varphi = f$. We show that $\bar{\alpha}$ and $\bar{f}$ are inverses of each other. To this end, see that

$$\bar{f} \circ \bar{\alpha}(g(X) + I) = \bar{f} \circ \alpha(g(X)) = \bar{f} \circ g(\frac{1}{a}) = (b_n + I)\bar{f}(\frac{1}{a^n}) + (b_{n-1} + I)\bar{f}(\frac{1}{a^{n-1}}) + \cdots (b_0 + I) \tag{1.52}$$

which is $(b_n + I)(X^n + I) + \cdots + (b_1 + I)(X + I) + b_0 + I = (b_nX^n + b_{n-1}X^{n-1} + \cdots + b_0) + I = g(X) + I$. Similarly, one can show that the $\bar{\alpha} \circ \bar{f}$. ■

1.8.1 Local Ring

Definition 1.89. A ring is called a *local ring* if it has a unique maximal ideal.

Of course, a trivial example is fields; the unique maximal ideal is $\{0\}$. In the ring of power series $k[[X]]$, (X) is the unique maximal ideal making $k[[X]]$ a local ring for a field k. $\mathbb{Z}/p^r\mathbb{Z}$ is a local ring with $p\mathbb{Z}/p^r\mathbb{Z}$ the unique maximal ideal.

For a more particular example, let $R \subseteq \mathbb{C}(X)$ be the subring of the field of complex rational functions defined as follows: $R = \{f(X)/g(X) \mid f(X), g(X) \in \mathbb{C}[X] \text{ and } g(0) \neq 0\}$. One can verify that this is a ring; it is the ring of rational functions from $\mathbb{C}$ to $\mathbb{C}$ which is defined at 0. Since such a function is continuous in its domain, it is locally define at 0. In fact, R is a local ring with the maximal ideal consisting of those rational functions h such that $h(0) = 0$.

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Proposition 1.90. *Let R be a ring and P be a prime ideal of R. Set $S = R \setminus P$. Then $S^{-1}R$ is a local ring with unique maximal ideal $PS^{-1}R = \{\frac{r}{s} \mid r \in P,\ s \in R \setminus P\}$.*

Proof. Note that it is enough to show that $PS^{-1}R$ is an ideal and $\alpha \in S^{-1}R \setminus PS^{-1}R$ is a unit. To show that $PS^{-1}R$ is an ideal, let $\frac{r_1}{s_1}, \frac{r_2}{s_2} \in PS^{-1}R$, so that $r_1, r_2 \in P$ and $s_1, s_2 \in R \setminus P$. Then $s_2r_1 + s_1r_2 \in P$ since P is and ideal, and $s_1s_2 \in R \setminus P$. Thus, $\frac{s_2r_1+s_1r_2}{s_1s_2} \in PS^{-1}R$. Also if $\frac{R}{s} \in S^{-1}R$, then $\frac{rr_1}{ss_1} \in PS^{-1}R$. Hence $PS^{-1}R$ is an ideal of $S^{-1}R$. Now let $\alpha = \frac{r}{s} \in S^{-1}R \setminus PS^{-1}R$. Since $\alpha \notin PS^{-1}R$, $r \notin P$. This means that $r \in R \setminus P = S$, so $\frac{s}{r} \in S^{-1}R$ is the multiplicative inverse of α. Thus, $S^{-1}R$ is a local ring with unique maximal ideal $PS^{-1}R$. ■

Proposition 1.91. *Let R be a ring $\mathfrak{m}$ be a maximal ideal of R. Then the following are equivalent:*

(i) R is a local ring.

(ii) The set of non-units of R form the ideal $\mathfrak{m}$.

(iii) $1 + x$ is a unit for all $x \in \mathfrak{m}$.

Proof. $(i) \implies (ii)$: Let J be the set of non-units of R. Note that $\mathfrak{m} \subseteq J$ since $\mathfrak{m}$ is a proper ideal. Let $x \in J$. Since x is not a unit, there exists a maximal ideal $\mathfrak{m}'$ containing x. Since R is a local ring, $\mathfrak{m}' = \mathfrak{m}$, so $x \in \mathfrak{m}$. Thus, $J \subseteq \mathfrak{m}$, so $J = \mathfrak{m}$.

$(ii) \implies (iii)$: If $x \in \mathfrak{m}$ and $1 + x$ is not a unit, then $1 + x \in \mathfrak{m}$. But then $1 = (1 + x) - x \in \mathfrak{m}$, contradicting the fact that $\mathfrak{m}$ is a proper ideal. Thus, $1 + x$ is a unit for all $x \in \mathfrak{m}$.

$(iii) \implies (i)$: Let $x \in \mathfrak{m}$. Then $ax \in \mathfrak{m}$ for all $a \in R$, and thus $1 + ax$ is a unit in R for all $a \in R$. Hence, $x \in \mathrm{Jac}(R)$. Finally, we have $\mathfrak{m} \subseteq \mathrm{Jac}(R)$ and so, $\mathfrak{m} = \mathrm{Jac}(R)$ since $\mathrm{Jac}(R) \subseteq \mathfrak{m}$ for any maximal ideal $\mathfrak{m}$. Thus, R is a local ring. ■

We look at some examples of using these propositions to construct local rings.

Example 1.92. Let $R = \mathbb{Z}$ and $P = (3)$. Then $S = \mathbb{Z} \setminus (3) = \{1, 2, 4, 5, 7, 8, \ldots\}$ and $S^{-1}R$ is the ring $\mathbb{Z}_{(3)} = \{\frac{a}{b} \mid a, b \in \mathbb{Z},\ 3 \nmid b\}$, which is a local ring with unique maximal ideal $\{\frac{a}{b} \mid a, b \in \mathbb{Z},\ 3 \nmid b,\ 3|a\}$.

1.9 Polynomials and Gauss Lemma

March 13th.

Gauss lemma can refer to several results; the one Gauss originally proved is the following: let $f(X) \in \mathbb{Z}[X]$ be a polynomial such that $p \nmid f(X)$ for all primes p. If $f(X) = g(X)h(X)$ in $\mathbb{Q}[X]$ then there exists a factorization $f(X) = G(X)H(X)$ in $\mathbb{Z}[X]$ such that $\deg(g) = \deg(G)$ and $\deg(h) = \deg(H)$. We will see this result as a corollary of several sequence of results, which are also called Gauss lemma.

Definition 1.93. Let R be a ring which admits a gcd function (for example, a UDF). For $f(X) \in R[X]$, we define the *content* of $f(X)$ as $c(f) = \gcd(a_0, a_1, \ldots, a_n)$ where $f(X) = a_n X^n + \cdots + a_1 X + a_0$. Note that $c = c(f)$ holds if and only if $uc = c(f)$ for all units $u \in R$. A polynomial $f(X)$ is called *primitive* if $c(f) = 1$.

As an example, $2X - 1$ is a primitive polynomial in $\mathbb{Z}[X]$ since $c(2X - 1) = 1$. On the other hand, $2X + 2$ is not a primitive polynomial in $\mathbb{Z}[X]$ since $c(2X + 2) = 2$. Note that if $f(X) \in R[X]$ is a non-zero polynomial, then $f(X) = c(f)f_0(X)$ for some primitive polynomial $f_0(X)$; indeed, we can take $f_0(X) = \frac{1}{c(f)} f(X)$.

The first version of Gauss lemma is the following.

Lemma 1.94 (Gauss lemma, version 1)**.** *Let R be a UFD and $f(x), g(x) \in R[X]$. Then $c(fg) = c(f)c(g)$.*

Lemma 1.95 (Gauss lemma, version 2)**.** *Let R be a UFD and $K = \mathrm{frac}(R)$ be its field of fractions. Let $f(X) \in R[X] \subseteq K[X]$. If $f(X) = g(X)h(X)$ for some $g(X), h(X) \in K[X]$, then $f(X)$ can be factored as $f(X) = G(X)H(X)$ for some $G(X), H(X) \in R[X]$ such that $\deg(G) = \deg(g)$ and $\deg(H) = \deg(h)$.*

One can easily see that the statement at the beginning of this section is a corollary of the second version of Gauss lemma. The following version of Gauss lemma, a corollary of the second version, is the most widely used version of Gauss lemma.

Corollary 1.96 (Gauss lemma, version 3)**.** *Let R be a UFD and $K = \mathrm{frac}(R)$ be its field of fractions. A primitive polynomial $f(X) \in R[X]$ is irreducible in $R[X]$ if and only if it is irreducible in $K[X]$.*

We first begin with the proof of this corollary.

Proof of Gauss lemma, version 3 from version 2. For the forward implication, let $f(X) \in R[X]$ be a primitive polynomial which is irreducible in $R[X]$, and suppose $f(X) = g(X)h(X)$ for some $g(X), h(X) \in K[X]$. By version 2 of Gauss lemma, there exist some $G(X), H(X) \in R[X]$ such that $f(X) = G(X)H(X)$ and $\deg(G) = \deg(g) \geq 1$ and $\deg(H) = \deg(h) \geq 1$. Hence, G and H are non-units in $R[X]$, contradicting the irreducibility of $f(X)$ in $R[X]$. For the converse implication, assume $f(X)$ is irreducible in $K[X]$, and suppose $f(X) = G(X)H(X)$ for some $G(X), H(X) \in R[X]$. Since $f(X)$ is irreducible in $K[X]$, either $G(X)$ or $H(X)$ is a constant polynomial; let's say $G(X)$ is a constant polynomial. Then $f(X) = cH(X)$ for some $c \in R$. Since $f(X)$ is primitive, we must have $c \mid c(f) = 1$, so c is a unit in R. Thus, $f(X)$ is irreducible in $R[X]$. ■

We now prove the second verion.

Proof of Gauss lemma, version 2 from version 1. Let $f(X) \in R[X]$ be a polynomial such that $f(X) = g(X)h(X)$ with $g(X), h(X) \in K[X]$. Note that since each coefficient of $g(X)$ and $h(X)$ is a fraction, we can find some β (possibly, the product of all denominators of coefficients of $g(X)$) and α (similarly pick α) such that $G_1(X) = \beta g(X)$ and $H_1(X) = \alpha h(X)$ are in $R[X]$. Then $\alpha\beta f(X) = G_1(X)H_1(X)$, so $c(\alpha\beta f(X)) = \alpha\beta c(f) = c(G_1)c(H_1)$. So $f(X) = c(f)F(X)$ where $F(X) \in R[X]$. Similarly, $G_1(X) = c(G_1)G(X)$ and $H_1(X) = c(H_1)H(X)$ for some $G, H \in R[X]$. Thus, $c(f)F(X) = c(G_1)c(H_1)G(X)H(X)$, so $F(X) = uG(X)H(X)$ where $u \in R$ is a unit. Hence, $uc(f)G(X), H(X) \in R[X]$ and $f(X) = c(F)F(X)$, with $\deg(uc(f)G(X)) = \deg(G(X)) = \deg(g(X))$ and $\deg(H(X)) = \deg(h(X))$. ■

Finally, we prove the first version of Gauss lemma.

Proof of Gauss lemma, version 1. Let $f(X), g(X) \in R[X]$, and $h(X) = f(X)g(X)$. Let $f(X) = c(f)F(X)$, $g(X) = c(g)G(X)$, $h(X) = c(h)H(X)$ where $F(X), G(X), H(X) \in R[X]$. Then $c(h)H(X) = c(f)c(g)F(X)G(X)$, so Note that $c(f)c(g)$ divides all coefficients of $h(X)$, so $c(f)c(g) \mid c(h)$. Let $c(f)c(g) = p_1 \cdots p_n$ where p_i's are irreducible in R and $c(h) = p_1 \cdots p_n q_1 \cdots q_m$ where q_j's are irreducible in R, and $m, n \geq 0$. If $m \neq 0$, then q_1 exists and $q_1 \mid c(h)$, so that $p_1 \cdots p_n q_1 \mid c(h)H(X) = c(f)c(g)F(X)G(X)$. Thus $q_1 \mid F(X)G(X)$ since $c(f)c(g) = p_1 \cdots p_n$. Let i_0 be the smallest index i such that $q_1 \nmid b_{i_0}$ and let j_0 be

the smallest index j such that $q_1 \nmid c_{j_0}$. Now look at the coefficient a of $X^{i_0+j_0}$ in $F(X)G(X)$. Then $q_1 \mid a$. But $a = b_{i_0}c_{j_0} + b_{i_0-1}c_{j_0+1} + \cdots + b_0c_{j_0+i_0} + b_{i_0+1}c_{j_0-1} + \cdots b_{i_0+j_0}c_0 = b_{i_0}c_{j_0} +$ (terms divisible by q_1). Since $q_1 \nmid b_{i_0}$ and $q_1 \nmid c_{j_0}$, $q_1 \nmid b_{i_0}c_{j_0}$, so $q_1 \nmid a$, a contradiction since q_1 is a prime. Thus, $m = 0$, so $c(h) = c(f)c(g)$. ■

Example 1.97. Let $f(X,Y) \in \mathbb{C}[X,Y]$ be $f(X,Y) = 2X^3Y + 3X^2 + 7XY + 4$. Then $f(X,Y)$ is also in $\mathbb{C}(X)[Y]$, where $\mathbb{C}(X)$ is the formal power series field in X over $\mathbb{C}$. Note that $f(X,Y) = (2X^3 + 7X)Y + (3X^2 + 4)$ is irreducible in $\mathbb{C}(X)[Y]$ since it is linear in Y. One can verify that $\gcd(2X^3 + 7X, 3X^2 + 4) = 1$ so $f(X,Y)$ is primitive in $\mathbb{C}[X][Y]$. By Gauss lemma, $f(X,Y)$ is irreducible in $\mathbb{C}[X][Y] = \mathbb{C}[X,Y]$.

Theorem 1.98. *Let R be a UFD. Then $R[X]$ is also a UFD.*

Proof. Let $f(X) \in R[X]$, and write $f(X) = c(f)F(X)$ for some primitive $F(X) \in R[X]$ and $c(f) \in R$. Since R is a UFD, we have $c(f) = p_1 \cdots p_r$ for some irreducibles $p_i \in R$. Let $K = \mathrm{frac}(R)$. Then $F(X) \in K[X]$. But $K[X]$ is a euclidean domain, and hence a UFD; hence, $F(X) = g_1(X) \cdots g_n(X)$ where $g_i(X) \in K[X]$ are irreducible polynomials. By the second version of Gauss lemma, $F(X) = G_1(X) \cdots G_n(X)$ where $G_i(X) \in R[X]$, and $G_i = \alpha_i g_i$ for some $\alpha_i \in K^\times$. Also, $1 = c(F) = c(G_1) \cdots c(G_n)$ implies G_i's are primitive polynomials and, so, irreducile polynomials in $K[X]$. Therefore, $G_i(X)$ are irreducible in $R[X]$ by version 3 of Gauss lemma. So $f(X) = p_1 \cdots p_r G_1(X) \cdots G_n(X)$. Suppose $f(X) = q_! \cdots q_t$ for q_i irreduciles in $R[X]$. Note that $c(f) = c(q_1) \cdots c(q_t)$ by the first version of Gauss lemma. Let $q_i(X) = c(q_i)Q_i(X)$ where $Q_i(X) \in R[x]$ are primitive. Since $q_1(X)$ is irreducible, either $c(q_i)$ or $Q_i(X)$ is a unit, that is, $q_i(X)$ is primitive or $q_i(X)$ is a constant polynomial. After reordering, we may assume $q_1, \ldots, q_s$ are constant polynomials and $q_{s+1}(X), \ldots, q_t(X)$ are non-constant primitive polynomials. So $c(f) = c(q_1) \cdots c(q_r) = q_1 \cdots q_s$ and q_i are irreducible constants in $R[X]$ and hence irreducible in R. Since R is a UFD, $r = s$ and p_i's are associate to q_i's after reordering. Thus, $f(X) = c(f)F(X) = q_{s+1}(X) \cdots q_t(X)(q_1 \cdots q_s)$ so $F(X) = q_{s+1}(X) \cdots q_t(X) = G_1(X) \cdots G_n(X)$ in $R[X] \subseteq K[X]$. Since $K[X]$ is a UFD, after reordering, $n = t - s$ and $G_i(X) = u_i q_{s+i}(X)$ for some units $u_i \in K^\times$. Of course, these may not be units in R; let $u_i = \frac{a_i}{b_i}$ for some $a_i, b_i \in R$ with $b_i \neq 0$. Then $b_i G_i(X) = a_i q_{s+i}(X)$ which implies $c(B_i G_i(X)) = c(a_i q_{s+i}(X))$ showing a_i and b_i are associates, or that $\frac{a_i}{b_i}$ is a unit in R. ■

Eisenstein's Criterion

March 16th.

Theorem 1.99. *Let $f(X) \in \mathbb{Z}[X]$ be a monic polynomial and p be a prime such that $p \mid a_i$ for all $i < n$ and $p^2 \nmid a_0$. Then $f(X)$ is irreducible in $\mathbb{Z}[X]$ (and hence in $\mathbb{Q}[X]$).*

Proof. Suppose $f(X) = g(X)h(X)$ for some $g(X), h(X) \in \mathbb{Z}[X]$. We may assume $g(X)$ and $h(X)$ are monic. Let $g(X) = X^m + b_{m-1}X^{m-1} + \cdots + b_0$ and $h(X) = X^l + c_{l-1}X^{l-1} + \cdots + c_0$. Let i_0 be the least index i such that $p \nmid b_{i_0}$, and j_0 be the least index j such that $p \nmid c_{j_0}$. Now consider $a_{i_0+j_0}$, the coefficient of $X^{i_0+j_0}$ in $f(X) = g(X)h(X)$. If $i_0 < m$ or $j_0 < l$, then $p \mid a_{i_0+j_0}$. We have $a_{i_0+j_0} = b_{i_0}c_{j_0} +$ (terms divisible by p), so $p \mid b_{i_0}c_{j_0}$. But then p must divide either b_{i_0} or c_{j_0}, contradicting the definition of i_0 and j_0. Thus, $i_0 = m$ and $j_0 = l$, so $p \mid b_i$ for all $i < m$ and $p \mid c_j$ for all $j < l$. Hence, $p^2 \mid a_0 = b_0c_0$, contradicting the assumption. Thus, $f(X)$ is irreducible in $\mathbb{Z}[X]$. ■

1.10 Noetherian Rings

Proposition 1.100. *Let R be a ring. The following are equivalent.*

1. *Every ideal of R is finitely generated.*
2. *Every increasing sequence of ideals stabilizes.*
3. *Let Ω be any non-empty collection of ideals of R. Then Ω has a maximal element under inclusion.*

Any ring satisfying any (all) of the three conclusions is termed a *noetherian ring*. Examples include any principal ideal domain, and fields. If $R_1, \ldots, R_n$ are noetherian, then so is their product. If R is noetherian and I is an ideal of R, then R/I is also noetherian. For a noetherian ring R and S a multiplicative subset, $S^{-1}R$ is also noetherian. The Hilbert basis theorem states that if R is noetherian, then so is $R[X]$.

Proof. 1. implies 2.: Let $I_0 \subseteq I_1 \subseteq I_2 \subseteq \cdots$ be an increasing sequence of ideals of R. Let $I = \bigcup_{n=0}^{\infty} I_n$, which is an ideal of R. Since I is finitely generated, we have $I = (x_1, \ldots, x_n)$ for some n with $x_i \in I$. Since $x_i \in I$, there exists N_i such that $x_i \in I_{N_i}$. Let $N = \max\{N_1, \ldots, N_n\}$. Then $x_1, \ldots, x_n \in I_N$, so $I = (x_1, \ldots, x_n) \subseteq I_N$. Since $I_N \subseteq I$, we have $I = I_N$. Hence, the sequence stabilizes at I_N.

2. implies 3.: Let $I_0 \in \Omega$. If I_0 is not maximal, there exists I_1 such that $I_0 \subsetneq I_1$ and $I_1 \in \Omega$. If I_1 is not maximal, there exists I_2 such that $I_1 \subsetneq I_2$ and $I_2 \in \Omega$. Continuing in this way, we get a strictly increasing sequence $I_0 \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$ of ideals in Ω, contradicting the assumption. Hence, there exists a maximal element in Ω.

3. implies 1.: Let $I \subseteq R$ be an ideal. Let $x_1 \in I$. If $I = (x_1)$, then I is finitely generated. Otherwise, there exists $x_2 \in I \backslash (x_1)$. If $I = (x_1, x_2)$, then I is finitely generated. Otherwise, there exists $x_3 \in I \backslash (x_1, x_2)$. Continuing in this way, we get a sequence of ideals $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$. If this process never terminates, then $\Omega = \{I_1, I_2, \ldots\}$ has no maximal element, contradicting the assumption. Hence, there exists n such that $I = (x_1, \ldots, x_n)$, so I is finitely generated. ■

March 20th.

We can prove the previously mentioned results about noetherian rings via the following small lemma.

Lemma 1.101.

- *If R is a noetherian ring and I is an ideal of R, then R/I is also noetherian.*
- *If R is a noetherian ring and S is a multiplicative subset of R, then $S^{-1}R$ is also noetherian.*
- *If $R_1, \ldots, R_n$ are noetherian rings, then so is $R_1 \times \cdots \times R_n$.*

Proof.

- Let $J \subseteq R/I$ be an ideal. Then $\tilde{J} := q^{-1}(J)$ is an ideal of R containing I, where $q : R \to R/I$ is the quotient map. Since R is noetherian, $\tilde{J} = (x_1, \ldots, x_n)$ for some $x_i \in R$. Then $J = q(\tilde{J}) = (q(x_1), \ldots, q(x_n))$, so J is finitely generated. Hence, R/I is noetherian.
- Let $J \subseteq S^{-1}R$ be an ideal. Then $I := \varphi^{-1}(J)$ is an ideal of R, where $\varphi : R \to S^{-1}R$ is the localization map. Also, $IS^{-1}R := \varphi(I)SR^{-1} = J$. Since R is noetherian, I is finitely generated, so $I = (x_1, \ldots, x_n)$ for some $x_i \in R$. Then $J = \varphi(I)S^{-1}R$ is generated by $\frac{x_1 1}{,} \ldots, \frac{x_n}{1}$. Let $\alpha \in J = \varphi(I)S^{-1}R$. Then $\alpha = \sum_{i=1}^{n} a_i b_i$ where $a_i \in S^{-1}R$ and $b_i \in \varphi(I)$. Since $b_i \in \varphi(I)$, there exists $y_i \in I$ such that $\varphi(y_i) = b_i$. So $y_i = \sum_{j=1}^{n} r_{ij} x_j$ for some $r_{ij} \in R$ since $I = (x_1, \ldots, x_n)$. Thus, $\alpha = \sum_{i=1}^{n} a_i b_i = \sum_{i=1}^{n} a_i \varphi(y_i) = \sum_{i=1}^{n} a_i \sum_{j=1}^{n} r_{ij} \varphi(x_j)$, so J is generated by $\frac{x_1}{1}, \ldots, \frac{x_n}{1}$.
- Let $J \subseteq R_1 \times \cdots \times R_n$ be an ideal. Then $J = I_1 \times \cdots \times I_n$ for ideals I_j of R_j for $j = 1, \ldots, n$. Since R_j is noetherian, $I_j = (x_{j1}, \ldots, x_{jm_j})$ for some $x_{jk} \in R_j$ and $m_j \geq 0$. Then J is generated by the elements $(x_{11}, 0, \ldots, 0), \ldots, (x_{1m_1}, 0, \ldots, 0), \ldots, (0, \ldots, 0, x_{n1}), \ldots, (0, \ldots, 0, x_{nm_n})$. Hence, $R_1 \times \cdots \times R_n$ is noetherian.

■

Example 1.102. As a counter-example to the point that a *countable* product of noetherian rings is noetherian, consider $R = \mathbb{Q}[X_1, X_2, \ldots]$, the polynomial ring in countably many variables over $\mathbb{Q}$. Look at the ideal $I = (X_1, X_2, \ldots)$, and suppose it were finitely generated by $f_1, \ldots, f_m \in \mathbb{Q}[X_1, X_2, \ldots]$. Let N be the largest index of variables appearing in $f_1, \ldots, f_m$. Then $X_{N+1} \in I$ cannot be generated by $f_1, \ldots, f_m$, a contradiction. Hence, I is not finitely generated, so R is not noetherian.

Theorem 1.103 (The *Hilbert basis theorem*)**.** *If R is a noetherian ring, then $R[X]$ is also noetherian.*

Proof. Let $I \subseteq R[X]$ be an ideal. Let $f_1 \in I$ be a non-zero polynomial of least degree possible. Consider the ideal $(f_1) \subseteq I$. If $I = (f_1)$, then I is finitely generated. Otherwise, let $f_2 \in I \setminus (f_1)$ be a non-zero polynomial of least degree possible. Consider the ideal $(f_1, f_2) \subseteq I$. If $I = (f_1, f_2)$, then I is finitely generated. Otherwise, we can continue this way to get a sequence of ideals $(f_1) \subsetneq (f_1, f_2) \subsetneq (f_1, f_2, f_3) \subsetneq \cdots$ in $R[X]$. Let a_i be the leading coefficient of $f_i(X)$ for $i \geq 1$. Then $a_1, a_2, \ldots$ are elements of R. Define the ideals $J_n = (a_1, \ldots, a_n)$ for $n \geq 1$. Note that $J_1 \subseteq J_2 \subseteq \cdots$ is an increasing sequence of ideals in R. Since R is noetherian, there exists N such that $J_N = J_{N+1} = \cdots$. We claim that $I = (f_1, \ldots, f_N)$. It is enough to show that f_{N+1} cannot exist. Suppose not; then $f_{N+1} = a_{N+1}X^n + (\text{lower degree terms})$, and $n = \deg(f_{N+1}) \geq \deg(f_i)$ for all $i = 1, \ldots, N$. $a_{N+1} \in J_N$ implies $a_{N+1} = b_1 a_1 + \cdots + b_N a_N$ for some $b_i \in R$. Now consider the polynomial

$$g(X) = f_{N+1} - \left(b_1 X^{n-\deg(f_1)} f_1 + b_2 X^{n-\deg(f_2)} f_2 + \cdots + b_N X^{n-\deg(f_N)} f_N\right); \tag{1.53}$$

then $g(X) \notin (f_1, \ldots, f_N)$ since $f_{N+1} \notin (f_1, \ldots, f_N)$ and $f_1, \ldots, f_N \in (f_1, \ldots, f_N)$. Also, $g(X) \in I$ since $f_1, \ldots, f_{N+1} \in I$. But $\deg(g) < \deg(f_{N+1}) = n$ since $g(X) = a_{n+1}X^n + (\text{lower degree terms}) - (a_1 b_1 X^n + a_2 b_2 X^n + \cdots + a_N b_N X^n + (\text{lower degree terms}))$. This contradicts the choice of f_{N+1} as a non-zero polynomial in $I \setminus (f_1, \ldots, f_N)$ of least degree possible. Hence, $I = (f_1, \ldots, f_N)$, so I is finitely generated. ■

As a small remark, if $R[X]$ is noetherian, then R is also noetherian since $R \cong R[X]/(X)$.

Chapter 2

MODULES

2.1 Abstracting Vector Spaces

Just as in the way that vector spaces are defined over fields, we can analogously define modules over rings. The theory of modules extends beyond commutative rings with unity to non-commutative rings as well, but we will restrict ourselves to the commutative case.

Definition 2.1. Let R be a ring. An *R-module* $(M, +, \cdot, R)$ where $+ : M \times M \to M$ is a binary operation and $\cdot : R \times M \to M$ is a function satisfying the following axioms.

- $(M, +)$ is an abelian group.
- $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ for all $r \in R$ and $m_1, m_2 \in M$.
- $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$ for all $r_1, r_2 \in R$ and $m \in M$.
- $1 \cdot m = m$ for all $m \in M$.
- $(r_1 r_2) \cdot m = r_1 \cdot (r_2 \cdot m)$ for all $r_1, r_2 \in R$ and $m \in M$.

Note that if R is a field, then an R-module is precisely a vector space over R.

Example 2.2. As a non-trivial example of a module, let $R = \mathbb{Z}$. Then a $\mathbb{Z}$-module is precisely an abelian group, since the axioms of a $\mathbb{Z}$-module reduce to the axioms of an abelian group; one can show this. Conversely, any abelian group can be regarded as a $\mathbb{Z}$-module by defining $n \cdot m = m + m + \cdots + m$ (n times) for $n > 0$, $0 \cdot m = 0$ for all $m \in M$, and $(-n) \cdot m = -(n \cdot m)$ for $n > 0$. More trivially, if R is a ring, then R itself is an R-module by the natural mappings. If I is an ideal of R, then I is an R-module by the natural mappings.

Example 2.3. Suppose $\varphi : R \to R'$ is a homomorphism of rings. Then R' is an R-module by defining $r \cdot r' = \varphi(r)r'$ for all $r \in R$ and $r' \in R'$. In particular, if φ is an inclusion map, then R' is an R-module by the natural mappings. More particularly, $S^{-1}R$ is an R-module for any multiplicatively closed subset S of R by the natural mappings.

Some trivial consequences of the axioms of a module can be inferred easily. For instance, $0 \cdot m = 0$ for all $m \in M$ since $0 \cdot m = (0 + 0) \cdot m = 0 \cdot m + 0 \cdot m$ implies $0 \cdot m = 0$. Similarly, $r \cdot 0 = 0$ for all $r \in R$. Also, $(-1) \cdot m = -m$ for all $m \in M$ since $0 = 0 \cdot m = (1 + (-1)) \cdot m = 1 \cdot m + (-1) \cdot m = m + (-1) \cdot m$ implies $(-1) \cdot m = -m$. More generally, $(-r) \cdot m = -(r \cdot m)$ for all $r \in R$ and $m \in M$.

Definition 2.4. Let M be an R-module. A non-empty subset $N \subseteq M$ is called an *R-submodule* of M if for all $x, y \in N$ and $r \in R$, $rx + y \in N$. That is, $(N, +|_{N \times N}, \cdot|_{R \times N}, R)$ is an R-module.

From the above, any ideal I of R is a submodule of the R-module R.

Proposition 2.5. *Let R be a ring and M be an R-module. Also let N be a submodule of M. Then M/N has a natural structure of an R-module defined by $R \times M/N \to M/N$ given by $(r, m+N) \mapsto r \cdot (m+N) := rm + N$ for all $r \in R$ and $m \in M$.*

This is called the *quotient module* of M by N.

Proof. First, we verify that the above definition is well-defined. Let $m+N = m'+N$ for some $m, m' \in M$. Then $m - m' \in N$, so $r(m-m') \in N$ since N is a submodule of M. Hence, $rm + N = rm' + N$, so the above definition is well-defined. It is left as an exercise to verify that the above definition indeed gives M/N the structure of an R-module, by checking the axioms of an R-module. ■

2.1.1 Linear Maps, and Isomorphisms

Definition 2.6. Let M, N be R-modules. A map $\varphi : M \to N$ is called *R-linear*, or *R-module homomorphism*, if $\varphi(x+y) = \varphi(x) + \varphi(y)$ for all $x, y \in M$ and $\varphi(rx) = r\varphi(x)$ for all $r \in R$ and $x \in M$.

Equivalently, φ is R-linear if and only if $\varphi(rx+y) = r\varphi(x) + \varphi(y)$ for all $r \in R$ and $x, y \in M$.

Example 2.7. If $N \subseteq M$ is an R-submodule, then the inclusion map $i : N \to M$ is R-linear, and injective. Also, the quotient map $q : M \to M/N$ given by $m \mapsto m + N$ is R-linear, and surjective.

An *R-module isomorphism* is simply a bijective R-linear map. If there exists an R-module isomorphism between M and N, then we say that M and N are isomorphic as R-modules.

Proposition 2.8. *Let $\varphi : M \to N$ be an R-linear map. Then $\ker\varphi = \{m \in M \mid \varphi(m) = 0\}$ is a submodule of M, and $\operatorname{Im}\varphi = \{\varphi(m) \mid m \in M\}$ is a submodule of N. M*

Proof. Let $x, y \in \ker\varphi$ and $r \in R$. Then $\varphi(rx+y) = r\varphi(x) + \varphi(y) = 0$, so $rx + y \in \ker\varphi$. Hence, $\ker\varphi$ is a submodule of M. Let $x, y \in \operatorname{Im}\varphi$ and $r \in R$. Then there exist $m_1, m_2 \in M$ such that $\varphi(m_1) = x$ and $\varphi(m_2) = y$. Then $\varphi(rm_1 + m_2) = r\varphi(m_1) + \varphi(m_2) = rx + y$, so $rx + y \in \operatorname{Im}\varphi$. Hence, $\operatorname{Im}\varphi$ is a submodule of N. ■

Theorem 2.9 (The first isomorphism theorem for modules)**.** *Let $\varphi : M \to N$ be an R-linear map. Let $K = \ker\varphi$ and $K' \subseteq K$ be an R-submodule. Then φ induces an R-linear map $\overline{\varphi} : M/K' \to N$ such that $\overline{\varphi}(m+K') = \varphi(m)$ for all $m \in M$. Moreover, $\overline{\varphi} \circ q = \varphi$, where $q : M \to M/K'$ is the quotient map. In particular, if $K' = K$, then $\overline{\varphi} : M/K \to N$ is an injective R-linear map, and if φ is surjective, then $\overline{\varphi} : M/K \to N$ is an R-module isomorphism.*

Proof. The map $\overline{\varphi}$ is well-defined since if $m+K' = m'+K'$ for some $m, m' \in M$, then $m - m' \in K' \subseteq K$, so $\varphi(m) = \varphi(m')$. Let $m+K', n+K' \in M/K'$ and $r \in R$. Then $\overline{\varphi}((m+K')+(n+K')) = \overline{\varphi}((m+n)+K') = \varphi(m+n) = \varphi(m)+\varphi(n) = \overline{\varphi}(m+K')+\overline{\varphi}(n+K')$ and $\overline{\varphi}(r(m+K')) = \overline{\varphi}(rm+K') = \varphi(rm) = r\varphi(m) = r\overline{\varphi}(m+K')$, so $\overline{\varphi}$ is R-linear. Also, for $m \in M$, $\overline{\varphi}(q(m)) = \overline{\varphi}(m+K') = \varphi(m)$, so $\overline{\varphi} \circ q = \varphi$. If $K' = K$, then $\ker\overline{\varphi} = \{m+K \mid m \in M, \varphi(m) = 0\} = \{m+K \mid \varphi(m) = 0\} = \{0+K\}$, so $\overline{\varphi}$ is injective. If φ is surjective, then $\overline{\varphi} \circ q = \varphi$ is surjective, so $\overline{\varphi}$ is surjective. Hence, $\overline{\varphi}$ is an R-module isomorphism. ■

Theorem 2.10 (The second isomorphism theorem for modules)**.** *Let M be an R-module, and let N_1, N_2 be R-submodules of M. Then $N_1 + N_2 = \{n_1 + n_2 \mid n_1 \in N_1, n_2 \in N_2\}$ is an R-submodule of M, and $N_1 \cap N_2$ is an R-submodule of M. Moreover,*

$$(N_1 + N_2)/N_2 \cong N_1/(N_1 \cap N_2) \tag{2.1}$$

Proof. Let $n_1, n_1' \in N_1$ and $n_2, n_2' \in N_2$, and let $r \in R$. Then $(n_1+n_2)+(n_1'+n_2') = (n_1+n_1')+(n_2+n_2') \in N_1+N_2$ and $r(n_1+n_2) = rn_1+rn_2 \in N_1+N_2$, so N_1+N_2 is an R-submodule of M. Let $x, y \in N_1 \cap N_2$ and $r \in R$. Then $x+y \in N_1$ and $x+y \in N_2$, so $x+y \in N_1 \cap N_2$. Also, $rx \in N_1$ and $rx \in N_2$, so $rx \in N_1 \cap N_2$. Hence, $N_1 \cap N_2$ is an R-submodule of M. Define $\varphi : N_1 \to (N_1+N_2)/N_2$ by $\varphi(n) = n+N_2$ for all $n \in N_1$. Then φ is well-defined since if $n, n' \in N_1$ and $\varphi(n) = \varphi(n')$, then $n - n' \in N_2$, so $n - n' \in N_1 \cap N_2$. Also, for all $n, n' \in N$ and $r \in R$, $\varphi(n+n') = (n+n')+N_2 = (n+N_2)+(n'+N_2) = \varphi(n)+\varphi(n')$ and $\varphi(rn) = rn+N_2 = r(n+N_2) = r\varphi(n)$, so φ is an R-linear map. Moreover, $\ker\varphi = N_1 \cap N_2$ since if $n \in N_1 \cap N_2$, then $\varphi(n) = n+N_2 = N_2$, so $n \in \ker\varphi$, and if $n \in \ker\varphi$, then $\varphi(n) = n+N_2 = N_2$, so $n \in N_2$, and since $n \in N_1$, $n \in N_1 \cap N_2$. Hence, $\ker\varphi = N_1 \cap N_2$. Also, $\operatorname{Im}\varphi = \{n+N_2 \mid n \in N_1\} = (N_1+N_2)/N_2$ since if $n \in N_1$, then $\varphi(n) = n+N_2 \in (N_1+N_2)/N_2$, and if $n+N_2 \in (N_1+N_2)/N_2$, then there exist $n_1 \in N_1$ and $n_2 \in N_2$ such that $n+N_2 = (n_1+n_2)+N_2 = n_1+N_2$, so $n_1 \in N_1$ and $\varphi(n_1) = n_1+N_2 = n+N_2$, so $n+N_2 \in \operatorname{Im}\varphi$. Hence, $\operatorname{Im}\varphi = (N_1+N_2)/N_2$. By the first isomorphism theorem for modules, $N_1/(N_1 \cap N_2) \cong (N_1+N_2)/N_2$. ■

Theorem 2.11 (The third isomorphism theorem for modules)**.** *Let M be an R-module, and let $K \subseteq N \subseteq M$ be R-submodules. Then N/K is an R-submodule of M/K, and $(M/K)/(N/K) \cong M/N$.*

The proof is left as an exercise, by defining $\varphi : M/K \to M/N$ by $\varphi(m+K) = m+N$ for all $m \in M$.

March 26th.

Proposition 2.12. *Let M and N be R-modules and $\varphi, \psi : M \to N$ be R-linear maps. Then $\varphi+\psi : M \to N$ defined by $(\varphi+\psi)(m) = \varphi(m)+\psi(m)$ for all $m \in M$ is an R-linear map, and $r\varphi : M \to N$ defined by $(r\varphi)(m) = r\varphi(m)$ for all $m \in M$ is an R-linear map for all $r \in R$.*

Proof. Let $m, m' \in M$ and $s \in R$. Then $(\varphi+\psi)(sm+m') = \varphi(sm+m')+\psi(sm+m') = \varphi(sm)+\varphi(m')+\psi(sm)+\psi(m') = (s\varphi(m)+s\psi(m))+(\varphi(m')+\psi(m')) = s(\varphi+\psi)(m)+(\varphi+\psi)(m')$, and $(r\varphi)(sm+m') = r\varphi(sm+m') = r(s\varphi(m)+\varphi(m')) = s(r\varphi)(m)+(r\varphi)(m')$, so $\varphi+\psi$ and $r\varphi$ are R-linear maps. ■

Thus, if we look at all R-linear maps from M to N, we in turn get another R-module.

Proposition 2.13. *Let M and N be R-modules. The set of all R-linear maps from M to N, denoted by $\operatorname{Hom}_R(M,N)$, is an R-module under the operations defined in the previous proposition.*

This proof is left as an exercise, by verifying the axioms of an R-module for $\operatorname{Hom}_R(M,N)$ under the operations defined in the previous proposition. We also define $\operatorname{End}_R(M) = \operatorname{Hom}_R(M,M)$, which is the set of all R-linear maps from M to itself. Note that $\operatorname{End}_R(M)$ is not just an R-module, but also a (possibly non-commutative) ring under composition of maps, with the identity map as the multiplicative identity. One can show also show that $\operatorname{Hom}_R(R,M) \cong M$ as R-modules, by defining $\varphi : \operatorname{Hom}_R(R,M) \to M$ by $\varphi(f) = f(1)$ for all $f \in \operatorname{Hom}_R(R,M)$, and showing that φ is an R-module isomorphism. Also, if M, N are R-modules, then $M \oplus N$ is an R-module under the natural operations.

Definition 2.14. Let M be a R-module, and $S \subseteq M$ be a subset. Then the submodule of M generated by S, denoted by $\langle S \rangle$, is the smallest submodule of M containing S. We call S a generating set for M if $\langle S \rangle = M$. S is called *linearly independent* if for all $n \in \mathbb{N}$, $r_1, \ldots, r_n \in R$ and $s_1, \ldots, s_n \in S$, $\sum_{i=1}^n r_i s_i = 0$ implies $r_1 s_1 = r_2 s_2 = \cdots = r_n s_n = 0$. A linearly independent generating set for M is called a *basis* for M.

Proposition 2.15. *For a subset S of an R-module M, $\langle S \rangle = \{\sum_{i=1}^n r_i s_i \mid n \in \mathbb{N}, r_1, \ldots, r_n \in R, s_1, \ldots, s_n \in S\}$.*

Definition 2.16. An R-module M is called a *free module* if M has a basis.

For example, if R is a ring, then R^n is a free R-module for all $n \in \mathbb{N}$, with the standard basis $e_1, \ldots, e_n$.

March 27th.

Proposition 2.17. *A finitely generated free R-module M is free if and only if $M \cong R^n$ for some $n \in \mathbb{N}$.*

Proof. If $M \cong \mathbb{R}^n$ via linear isomorphism $\varphi : \mathbb{R}^n \to M$, then set $x_i := \varphi(e_i)$ for all $1 \leq i \leq n$. If $\sum_{i=1}^n r_i x_i = 0$ for some $r_1, \ldots, r_n \in R$, then $\sum_{i=1}^n r_i e_i = \varphi^{-1}(\sum_{i=1}^n r_i x_i) = \sum_{i=1}^n r_i \varphi^{-1}(x_i) = \varphi^{-1}(0) = 0$, so $\sum_{i=1}^n r_i e_i = 0$, so $r_1 e_1 = r_2 e_2 = \cdots = r_n e_n = 0$, so $r_1 x_1 = r_2 x_2 = \cdots = r_n x_n = 0$. Also, for $m \in M$, there exist $r_1, \ldots, r_n \in R$ such that $\varphi(\sum_{i=1}^n r_i e_i) = m$, so $\sum_{i=1}^n r_i x_i = m$. Hence, $x_1, \ldots, x_n$ is a basis for M. Conversely, suppose M is a finitely generated free R-module. Let $\{y_1, y_2, \ldots, y_m\}$ be a finitely generating set of M and let S be a basis of M. Then for each $1 \leq i \leq m$, we have $y_i = \sum_{j=1}^{N_i} r_j x_{ij}$ for some $N_i \in \mathbb{N}$, $r_1, \ldots, r_{N_i} \in R$ and $x_{i1}, \ldots, x_{iN_i} \in S$. So, $\{x_{ij} \mid 1 \leq i \leq m,\ 1 \leq j \leq N_i\}$ is finite and generates M. Hence S is finite, so $S = \{x_1, \ldots, x_n\}$ for some $n \in \mathbb{N}$. Define $\varphi : R^n \to M$ by $\varphi(r_1, \ldots, r_n) = \sum_{i=1}^n r_i x_i$ for all $r_1, \ldots, r_n \in R$. Then φ is well-defined since if $(r_1, \ldots, r_n) = (r'_1, \ldots, r'_n)$ for some $r_1, \ldots, r_n, r'_1, \ldots, r'_n \in R$, then $r_i = r'_i$ for all $1 \leq i \leq n$, so $\sum_{i=1}^n r_i x_i = \sum_{i=1}^n r'_i x_i$. Also, for all $(r_1, \ldots, r_n), (s_1, \ldots, s_n) \in R^n$ and $t \in R$, $\varphi((r_1, \ldots, r_n) + (s_1, \ldots, s_n)) = \varphi(r_1 + s_1, \ldots, r_n + s_n) = \sum_{i=1}^n (r_i + s_i) x_i = \sum_{i=1}^n r_i x_i + \sum_{i=1}^n s_i x_i = \varphi(r_1, \ldots, r_n) + \varphi(s_1, \ldots, s_n)$ and $\varphi(t(r_1, \ldots, r_n)) = \varphi(tr_1, \ldots, tr_n) = t \sum_{i=1}^n r_i x_i = t\varphi(r_1, \ldots, r_n)$, so φ is an R-linear map. Moreover, $\ker \varphi = 0$ since if $\varphi(r_1, \ldots, r_n) = 0$ for some $r_1, \ldots, r_n \in R$, then $\sum_{i=1}^n r_i x_i = 0$, so $r_1 x_1 = r_2 x_2 = \cdots = r_n x_n = 0$, so $r_1 = r_2 = \cdots = r_n = 0$. Also, $\operatorname{Im} \varphi = M$ since for all $m \in M$, there exist $r_1, \ldots, r_n \in R$ such that $m = \sum_{i=1}^n r_i x_i$, so $\varphi(r_1, \ldots, r_n) = m$. Hence, φ is an R-module isomorphism, so $M \cong R^n$. ■

Proposition 2.18. *Let M and N be R-modules. Then, giving an R-linear map $\varphi : M \to N$ is equivalent to giving an element of N for each element of a basis of M. More precisely, if M is a free R-module with basis S, then for any function $\bar{\varphi} : S \to N$, there exists a unique R-linear map $\varphi : M \to N$ such that $\varphi(s) = \bar{\varphi}(s)$ for all $s \in S$.*

Proof. Given an R-linear map, $\bar{\varphi} := \varphi|_S : S \to N$ is a function. Conversely, given a function $\bar{\varphi} : S \to N$, define $\varphi : M \to N$ by $\varphi(\sum_{i=1}^n r_i s_i) = \sum_{i=1}^n r_i \bar{\varphi}(s_i)$ for all $n \in \mathbb{N}$, $r_1, \ldots, r_n \in R$ and $s_1, \ldots, s_n \in S$. It is simple to check that φ is well-defined and R-linear, and that $\varphi(s) = \bar{\varphi}(s)$ for all $s \in S$. Uniqueness also follows trivially. ■

Proposition 2.19. *Let M be an R-module and $I \subseteq R$ be an ideal. Then $IM = \{\sum_{i=1}^n x_i m_i \mid n \geq 1,\ x_1, \ldots, x_n \in I,\ m_1, \ldots, m_n \in M\}$ is an R-submodule of M. Moreover, M/IM is an R/I-module via the operations defined by $(r + I)(m + IM) = rm + IM$ for all $r \in R$ and $m \in M$.*

Before we being the proof, recall that via the quotient map $q : R \to R/I$, every module over R/I can be regarded as a module over R by defining $r \cdot m = q(r)m$ for all $r \in R$ and $m \in M$. The converse, however, is not true in general; that is, not every R-module can be regarded as an R/I-module.

Proof. Note that $IM \subseteq M$ is clear. Let $\alpha_1, \alpha_2 \in IM$ and $r \in R$. Then $\alpha_1 = x_1 m_1 + \cdots + x_r m_r$ and $\alpha_2 = y_1 n_1 + \cdots + y_s n_s$ for some $r, s \in \mathbb{N}$, $x_1, \ldots, x_r, y_1, \ldots, y_s \in I$ and $m_1, \ldots, m_r, n_1, \ldots, n_s \in M$. Then $r\alpha_1 + \alpha_2 = r(x_1 m_1 + \cdots + x_r m_r) + (y_1 n_1 + \cdots + y_s n_s) = (rx_1)m_1 + \cdots + (rx_r)m_r + y_1 n_1 + \cdots + y_s n_s$, so $r\alpha_1 + \alpha_2 \in IM$. Hence, IM is an R-submodule of M. Thus, the quotient module M/IM is well-defined, and an R-module under via $r \cdot (m + IM) = rm + IM$ for all $r \in R$ and $m \in M$. Moreover, for $r, r' \in R$, $m, m' \in M$ and $x \in I$, we have $(r + I)((r' + I)(m + IM)) = (r + I)(r'm + IM) = rr'm + IM$ and $((r + I)(r' + I))(m + IM) = (rr' + I)(m + IM) = rr'm + IM$, so the above definition indeed gives M/IM the structure of an R/I-module. ■

Example 2.20. Note that M being a finitely generated R-module does not imply that an R-submodule of M is finitely generated. For instance, let $R = \mathbb{Q}[X_1, X_2, \ldots]$ be the polynomial ring in countably many variables over $\mathbb{Q}$, and let $M = R$. Then M is a finitely generated R-module since $M = R \cdot 1$. However, the ideal $I = (X_1, X_2, \ldots)$ of R is not finitely generated, so I is an R-submodule of M that is not finitely generated.

2.1.2 Noetherian Modules

Proposition 2.21. *Let M be an R-module. Then the following are equivalent.*

1. *Every submodule of M is finitely generated.*
2. *Every ascending chain of submodules of M stabilizes.*
3. *Every non-empty set of submodules of M has a maximal element under inclusion.*

Such a module is called a *noetherian module*.

Proof.

1. Let $N = \bigcup_{i\geq 1} M_i$, for some ascending chain $M_1 \subseteq M_2 \subseteq \cdots$ of submodules of M. Then N is a submodule of M, so N is finitely generated by, say, $m_1, \ldots, m_k$. Take $n \in \mathbb{N}$ such that $m_1, \ldots, m_k \in M_n$. Then $M_n \subseteq M_{n+1} \subseteq \cdots \subseteq N$, so $M_n = M_{n+1} = \cdots = N$. Hence, the ascending chain stabilizes.
2. Let Ω be a non-empty set of submodules of M. If Ω has no maximal element, then for all $N \in \Omega$, there exists $N' \in \Omega$ such that $N \subsetneq N'$. Hence, we can construct an infinite ascending chain $N_1 \subsetneq N_2 \subsetneq \cdots$ of submodules of M by choosing $N_1 \in \Omega$ arbitrarily and choosing $N_{i+1} \in \Omega$ such that $N_i \subsetneq N_{i+1}$ for all $i \geq 1$. This contradicts the assumption that every ascending chain of submodules of M stabilizes. Hence, Ω has a maximal element.
3. Let N be a submodule of M. Let $\Omega = \{N' \subseteq N \mid N' \text{ is a submodule of } M\}$. Then Ω is non-empty since $0 \in \Omega$. Hence, Ω has a maximal element, say N_0. We claim that $N_0 = N$. If not, then there exists $m \in N \setminus N_0$. Then $N_0 + Rm$ is a submodule of M containing N_0 properly, so $N_0 + Rm \in \Omega$, contradicting the maximality of N_0. Hence, $N_0 = N$, so N is finitely generated by the maximal element of Ω.

■

Proposition 2.22. *Let M be a finitely generated R-module, and let $N \subseteq M$ be a submodule. Then M/N is finitely generated.*

Proof. Let $m_1, \ldots, m_k$ be such that $M = Rm_1 + \cdots + Rm_k$. For $\alpha = m + N \in M/N$, there exist $r_1, \ldots, r_k \in R$ such that $m = r_1m_1 + \cdots + r_km_k$, so $\alpha = m + N = r_1m_1 + \cdots + r_km_k + N = r_1(m_1 + N) + \cdots + r_k(m_k + N)$, so $M/N = R(m_1 + N) + \cdots + R(m_k + N)$, so M/N is finitely generated. ■

Proposition 2.23. *Let M be a noetherian R-module and $N \subseteq M$ be a submodule. Then M/N is noetherian.*

Proof. Let $K \subseteq M/N$ be an R-submodule of M/N. Then $\tilde{K} = q^{-1}(K)$ is a submodule of M containing N, where $q : M \to M/N$ is the quotient map. Since M is noetherian, $\tilde{K}$ is finitely generated. By the previous proposition, $K = \tilde{K}/N$ is finitely generated. Hence, M/N is noetherian. ■

Theorem 2.24. *Let R be a noetherian ring. If M is finitely generated as an R-module, then M is a noetherian module.*

Proof. Let $x_1, \ldots, x_n \in M$ be a generating set for M. Then $\varphi : R^n \to M$ extends to an R-linear sujrective map. By the first isomorphism theorem, $M \cong R^n / \ker \varphi$. Thus it is enough to prove that R^n is noetherian, since the previous proposition implies that a quotient of a noetherian module is noetherian. We shall prove this by induction. For $n = 1$, it is trivially true since an R-submodule of R is an ideal of R, and R is noetherian so ideals of R are finitely generated. Now assume R^{n-1} is noetherian. Look at the inclusion map $i : R^{n-1} \to R^n$ defined by $i(r_1, \ldots, r_{n-1}) = (r_1, \ldots, r_{n-1}, 0)$ for all $r_1, \ldots, r_{n-1} \in R$, and the projection map $\pi : R^n \to R$ defined by $\pi(r_1, \ldots, r_n) = r_n$ for all $r_1, \ldots, r_n \in R$. Then i is an injective R-linear map, and π is a surjective R-linear map. So, $\ker \pi \cong R^{n-1}$. Let $M_1 = M \bigcap \ker \pi$, which is isomorphic to an R-submodule of R^{n-1}. Hence, M_1 is finitely generated, by the induciton hypothesis, by, say, $\{y_1, \ldots, y_r\} \subseteq M_1$. Also, $\pi(M) \subseteq R$ is an R-submodule of R, that is, an ideal of R. So $\pi(M)$ is finitely generated by $\{z_1, \ldots, z_s\} \subseteq \pi(M)$. Let $\tilde{z}_i$ be such that $\pi(\tilde{z}_i) = z_i$ for all $1 \leq i \leq s$. We claim

that $\{y_1, \ldots, y_r, \tilde{z}_1, \ldots, \tilde{z}_s\}$ generates M. Let $\alpha \in M$. Then $\pi(\alpha) = \sum_{i=1}^s r_i z_i$ for some $r_1, \ldots, r_s \in R$, so $\pi(\alpha - \sum_{i=1}^s r_i \tilde{z}_i) = 0$, so $\alpha - \sum_{i=1}^s r_i \tilde{z}_i \in M \cap \ker \pi = M_1$, so $\alpha - \sum_{i=1}^s r_i \tilde{z}_i = \sum_{j=1}^r t_j y_j$ for some $t_1, \ldots, t_r \in R$, so $\alpha = \sum_{j=1}^r t_j y_j + \sum_{i=1}^s r_i \tilde{z}_i$, so $\{y_1, \ldots, y_r, \tilde{z}_1, \ldots, \tilde{z}_s\}$ generates M. Hence, M is noetherian. ■

2.2 Localization

March 30th.

Definition 2.25. Let R be a ring and $S \subseteq R$ be a multiplicative subset, and let M be an R-module. On $S \times M$, we define a relation $(s_1, m_1) \sim (s_2, m_2)$ if there exists $t \in S$ such that $t(s_2 m_1 - s_1 m_2) = 0$. It is simple to check that $\sim$ is an equivalence relation. Define $S^{-1}M := S \times M/\sim$; the equivalence classes of $S^{-1}M$ are denoted by $\frac{m}{s}$ for $m \in M$ and $s \in S$. $S^{-1}M$ has a natural $S^{-1}R$-module structure given by $\frac{m}{s} + \frac{n}{t} = \frac{tm+sn}{st}$ and $\frac{r}{s} \cdot \frac{m}{t} = \frac{rm}{st}$ for all $m, n \in M$, $r \in R$ and $s, t \in S$. We call $S^{-1}M$ the localization of M at S. These are well-defined and satisfy the axioms of an $S^{-1}R$-module. In particular, $S^{-1}M$ is also an R-module via $r \cdot \frac{m}{s} = \frac{rm}{s}$ for all $r \in R$, $m \in M$ and $s \in S$. Moreover, $\varphi : M \to S^{-1}M$ defined by $m \mapsto \frac{m}{1}$ for all $m \in M$ is an R-linear map.

All of these are straightforward to verify and mimic the construction of the localization of a ring.

Example 2.26. One can simply take $M = R$, and $S^{-1}M$ is just the localization of R at S, and works as an $S^{-1}R$-module under the natural operations.

Example 2.27. Let $R = \mathbb{Z}$ and $M = \mathbb{Z} \oplus \mathbb{Z}/15\mathbb{Z}$. Taking $S = \mathbb{Z} \setminus \{0\}$, we have $S^{-1}M = S^{-1}\mathbb{Z} \oplus S^{-1}(\mathbb{Z}/15\mathbb{Z}) = \mathbb{Q} \oplus 0$; note that $S^{-1}(\mathbb{Z}/15\mathbb{Z})$ is trivial since $15 \in S$ and $15m = 0$ for all $m \in \mathbb{Z}/15\mathbb{Z}$. Thus, $S^{-1}M = \mathbb{Q}$. Let us now take $S = \{1, 3, 9, 27, \ldots\}$. Then

$$S^{-1}M = S^{-1}\mathbb{Z} \oplus S^{-1}(\mathbb{Z}/3\mathbb{Z}) \oplus S^{-1}(\mathbb{Z}/5\mathbb{Z}) = \mathbb{Z}[\frac{1}{3}] \oplus 0 \oplus \mathbb{Z}/5\mathbb{Z} = \mathbb{Z}[\frac{1}{3}] \oplus \mathbb{Z}/5\mathbb{Z}. \tag{2.2}$$

This is how we work in general; if M is a direct sum of modules, then we can localize each component and take the direct sum of the localizations to get the localization of M. If $S = \{1, 15, 15^2, 15^3, \ldots\}$, then show that $S^{-1}M = \mathbb{Z}[\frac{1}{15}]$.

In the above, we used some basic properties of localization without explicit proof. These include that $S^{-1}(M \oplus N) = S^{-1}M \oplus S^{-1}N$, $S^{-1}(M/N) = S^{-1}M/S^{-1}N$, and $S^{-1}(R/I) = S^{-1}R/S^{-1}I$ for an ideal I of R.

April 2nd.

Proposition 2.28. *If $N \subseteq M$ is an R-submodule, then $S^{-1}N$ is an R-submodule of $S^{-1}M$, and $S^{-1}(M/N) \cong S^{-1}M/S^{-1}N$ as $S^{-1}R$-modules.*

Proof. Look at the map $\alpha : S^{-1}M \to S^{-1}(M/N)$ defined by $\frac{m}{s} \mapsto \frac{m+N}{s}$ for all $m \in M$ and $s \in S$. TO show this is well-defined, let $\frac{m}{s} = \frac{m'}{s'}$ for some $m, m' \in M$ and $s, s' \in S$. Then there exists $t \in S$ such that $t(s'm - sm') = 0$, which implies that $t(s'm + N - sm' - N) = N$ in M/N, so $\frac{m+N}{s} = \frac{m'+N}{s'}$. It is simple to check that α is an $S^{-1}R$-linear map, and is surjective. Moreover,

$$\ker \alpha = \{\frac{m}{s} \mid \frac{m+N}{s} = N\} = \{\frac{m}{s} \mid um \in N \text{ for some } u \in S\} = S^{-1}N \tag{2.3}$$

where last equality follows from the definition of $S^{-1}N$. Hence, by the first isomorphism theorem, $S^{-1}M/S^{-1}N \cong S^{-1}(M/N)$ as $S^{-1}R$-modules. ■

2.2.1 Rank

Definition 2.29. Let R be an integral domain, and M an R-module. The *rank* of M, as an R-module, is defined as $\operatorname{rank} M = \dim_{S^{-1}R}(S^{-1}M)$, where $S = R \setminus \{0\}$. Note that $S^{-1}R$ is the field of fractions of R, so $S^{-1}M$ is a vector space over $S^{-1}R$, so $\dim_{S^{-1}R}(S^{-1}M)$ is well-defined.

Example 2.30. Let $R = \mathbb{Z}$ and $M = \mathbb{Z}^2 \oplus \mathbb{Q} \oplus \mathbb{Z}/15\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$. Then $S^{-1}M = \mathbb{Q}^2 \oplus \mathbb{Q} \oplus 0 \oplus 0$, so $\operatorname{rank} M = 3$. If we let $M = n\mathbb{Z}$ instead, then $S^{-1}M = \mathbb{Q}$, so $\operatorname{rank} M = 1$. This is also easy to interpret since $n\mathbb{Z}$ is isomorphic to $\mathbb{Z}$ as a $\mathbb{Z}$-module, so $n\mathbb{Z}$ has the same rank as $\mathbb{Z}$, which is 1.

Example 2.31. Let $R = \mathbb{Q}[X, Y]$ and $I = (X, Y)$ be the ideal of R generated by X and Y. Then $S^{-1}I = S^{-1}R$, so $\operatorname{rank} I = 1$.

We now come to the universal property of localization, which is a very important property that characterizes the localization of a module up to isomorphism.

Theorem 2.32 (The *universal property of localization*)**.** *Let M be an R-module and $S \subseteq R$ be a multiplicative subset. Moreover, let $\varphi : M \to S^{-1}M$ be the R-linear localization map, N be an $S^{-1}R$-module, and $\theta : M \to N$ be an R-linear map. Then there exists a unique $S^{-1}R$-linear map $\tilde{\theta} : S^{-1}M \to N$ such that $\tilde{\theta} \circ \varphi = \theta$.*

Proof. Define $\tilde{\theta} : S^{-1}M \to N$ by $\tilde{\theta}(\frac{m}{s}) = \frac{1}{s}\theta(m)$ for all $m \in M$ and $s \in S$. It may be verified that $\tilde{\theta}$ is well-defined and $S^{-1}R$-linear, and that $\tilde{\theta} \circ \varphi = \theta$. Uniqueness is also left as an exercise. ■

2.2.2 Torsion

Definition 2.33. Let M be an R-module. Define

$$T(M) = \{m \in M \mid rm = 0 \text{ for some } r \in R \setminus \{0\}\} \subseteq M. \tag{2.4}$$

$T(M)$ here is called the *torsion* of M. Note that if R is an integral domain, then $T(M)$ is an R-submodule of M.

We show the second statement.

Proof. Let $m_1, m_2 \in T(M)$. Then there exist $r_1, r_2 \in R \setminus \{0\}$ such that $r_1 m_1 = 0$ and $r_2 m_2 = 0$. Let $r = r_1 r_2$, which is non-zero since R is an integral domain. Then $r(m_1 + m_2) = r_1(r_2 m_1) + r_2(r_1 m_2) = 0$, so $m_1 + m_2 \in T(M)$. Also, for $r' \in R$, we have $r'(rm_1) = (rr')m_1 = 0$, so $r'm_1 \in T(M)$. Hence, $T(M)$ is an R-submodule of M. ■

Example 2.34. Let $R = \mathbb{Z}/6\mathbb{Z}$, with $M = \mathbb{Z}/6\mathbb{Z}$. Then $T(M) = \{0, 2, 3, 4\}$, which is not a submodule of M since $2 + 3 = 5 \notin T(M)$.

Definition 2.35. An R-module M is called *torsion free* if $T(M) = 0$. M is called a *torsion module* if $T(M) = M$.

Proposition 2.36. *Suppose R is an integral domain, and M an R-module. Then $M/T(M)$ is torsion free.*

April 3rd.

Proof. Let $m + T(M) \in M/T(M)$ be such that $r(m + T(M)) = T(M)$ for some $r \in R \setminus \{0\}$. Then $rm \in T(M)$, so there exists $r' \in R \setminus \{0\}$ such that $r'(rm) = (rr')m = 0$, so $m \in T(M)$, so $m + T(M) = T(M)$. Hence, $M/T(M)$ is torsion free. ■

Proposition 2.37. *Let R be integral domain, and M be a torsion R-module. Then* $\operatorname{rank} M = 0$.

Proof. Let $S = R \setminus \{0\}$. Then $S^{-1}M = 0$ since for all $\frac{m}{s} \in S^{-1}M$, we have $rm = 0$ for some $r \in R \setminus \{0\}$, so $\frac{m}{s} = \frac{rm}{rs} = 0$. Hence, $\operatorname{rank} M = \dim_{S^{-1}R}(S^{-1}M) = 0$. ■

Proposition 2.38. $\operatorname{rank}(M \oplus N) = \operatorname{rank} M + \operatorname{rank} N$ *for all R-modules M and N.*

The proof is left as a straightforward exercise.

2.3 Structure Theorem

Theorem 2.39 (The *structure theorem for finitely generated modules over a PID*)**.** *Let R be a principal ideal domain, and M be a finitely generated R-module. Then*

$$M \cong R^k \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_n) \tag{2.5}$$

for some $k, n \geq 0$ and $a_1, \ldots, a_n \in R$ such that $a_1 \mid a_2 \mid \cdots \mid a_n$. Note that $k = 0$ if and only if M is a torsion module, and $n = 0$ if and only if M is torsion free. Moreover, the above decomposition is unique up to isomorphism.

Interesting corollaries can be drawn.

Corollary 2.40. *Let R be a principal ideal domain, and M be a finitely generated torsion free R-module. Then M is a free R-module.*

Theorem 2.41 (The *structure theorem for finitely generated abelian groups*)**.** *Let G be a finitely generated abelian group. Then*

$$G \cong \mathbb{Z}^k \oplus \mathbb{Z}/n_1\mathbb{Z} \oplus \mathbb{Z}/n_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/n_m\mathbb{Z} \tag{2.6}$$

for some $k, m \geq 0$ and $n_1, \ldots, n_m \in \mathbb{N}$ such that $n_1 \mid n_2 \mid \cdots \mid n_m$.

Proof. This is a direct consequence of the previous theorem, since a finitely generated abelian group is just a finitely generated $\mathbb{Z}$-module, and $\mathbb{Z}$ is a principal ideal domain. ■

Example 2.42. Let $R = \mathbb{Z}$ and $M = \mathbb{Q}$. This is not finitely generated as a $\mathbb{Z}$-module. Note that $\operatorname{rank} \mathbb{Q} = 1$ since $S^{-1}\mathbb{Q} = \mathbb{Q}$ as a $\mathbb{Q}$-vector space, so $\dim_{\mathbb{Q}}(S^{-1}\mathbb{Q}) = 1$. However, $\mathbb{Q}$ is not singly generated; let $\frac{p}{q}, \frac{r}{s} \in \mathbb{Q}$. Then $rq\frac{p}{q} - ps\frac{r}{s} = 0$, so any set of length 2 is linearly dependent.

Proposition 2.43. *Let R be a principal ideal domain, and F be a free module of rank n. Also let N be a submodule of F. Then N is also free and of rank $m \leq n$. Moreover, there is a basis $\{x_1, x_2, \ldots, x_n\}$ of F and elements $a_1, a_2, \ldots, a_m \in R$ such that $\{a_1x_1, a_2x_2, \ldots, a_mx_m\}$ is a basis of N and $a_1 \mid a_2 \mid \cdots \mid a_m$.*

Proof. This is trivial if $N = 0$, so assume $N \neq 0$. Let $\Sigma = \{\varphi(N) \mid \varphi : F \to R \text{ is an } R\text{-linear map}\}$. Then Σ is a non-empty collection of R-ideals, so Σ has a maximal element, say $I_1 = (a_1) = R$, with $a_1 \neq 0$. Also, $(a_1) = \nu(N)$ for some R-linear $\nu : F \to R$. There exists $x \in N$ such that $a_1 = \nu(x)$. We claim that $a_1 \mid \varphi(x)$ for all R-linear maps $\varphi : F \to R$. If not, then let $d = \gcd(a_1, \varphi(x))$ so that $d = r_1a_1 + r_2\varphi(x)$ for

some $r_1, r_2 \in R$. Consider $(r_1\nu + r_2\varphi) : F \to R$. Then $(r_1\nu + r_2\varphi)(x) = r_1a_1 + r_2\varphi(x) = d$, which implies $(r_1\nu + r_2\varphi)(N) \supseteq (d) \supsetneq (a_1)$, contradicting the maximality of (a_1).

In particular, $a_1 \mid \pi_i \circ \psi(x)$, where $\psi : F \to R^n$ is an isomorphism and $\pi_i : R^n \to R$ is the projection map defined by $\pi_i(r_1, \dots, r_n) = r_i$ for all $r_1, \dots, r_n \in R$. Let $\psi(x) = (b_1, \dots, b_n)$. Then $a_1 \mid b_i$ for all $1 \le i \le n$, so there exists $x_1 \in F$ such that $\psi(x_1) = (\frac{b_1}{a_1}, \dots, \frac{b_n}{a_1}) \in R^n$, and $a_1x_1 = x$ (possible since $\psi(a_1x_1) = (b_1, \dots, b_n)$). We now claim that $F = Rx_1 \oplus \ker\nu$ and $N = Ra_1x_1 \oplus (N \cap \ker\nu)$. To this end, let $y \in F$. Then $y = \nu(y)x_1 + (y - \nu(y)x_1)$, where $\nu(y)x_1 \in Rx_1$ and $y - \nu(y)x_1 \in \ker\nu$ (since $\nu(a_1x_1) = \nu(x)$, $\nu(x_1) = 1$). Now let $y \in Rx_1 \cap \ker\nu$. Then $y = rx_1$ for some $r \in R$, and $0 = \nu(y) = r\nu(x_1) = r$, so $y = 0$. Hence, $F = Rx_1 \oplus \ker\nu$. A similar argument shows that $N = Ra_1x_1 \oplus (N \cap \ker\nu)$. Thus, $\operatorname{rank} N = \operatorname{rank}(Ra_1x_1) + \operatorname{rank}(N \cap \ker\nu)$ and $\operatorname{rank} F = \operatorname{rank}(Rx_1) + \operatorname{rank}(\ker\nu)$. Note that $\ker\nu \cap N$ is an R-submodule of F and of rank strictly less than $\operatorname{rank} N$. Hence, by induction on the rank of N, $\ker\nu \cap N$ is free and of rank $\operatorname{rank}(N) - 1 = m - 1$. But $N = Ra_1x_1 \oplus (N \cap \ker\nu) \cong R^m$. Therefore, N is free and of rank m.

Now induct on the rank of F; $\ker\nu$ is an R-submodule of F and of rank $n - 1$. Also, it is free by what we just proved. Hence, $\ker\nu \cong R^{n-1}$. $N \cap \ker\nu \subseteq N$ is an R-submodule of N, so by the induction hypothesis, there exists a basis $\{x_2, \dots, x_n\}$ of $\ker\nu$ and elements $a_2, \dots, a_m \in R$ such that $a2 \mid \cdots \mid a_m$ and $\{a_2x_2, \dots, a_mx_m\}$ is a basis of $N \cap \ker\nu$. By the claim above, $\{x_1, \dots, x_n\}$ is a basis of F and $\{a_1x_1, a_2x_2, \dots, a_mx_m\}$ is a basis of N. We now wish to show that $a_1 \mid a_2$.

With Σ defined as beofre, and $(a_1) = \nu(N)$, we have $(a_2) = \nu_2(N \cap \ker\nu)$ for some R-linear $\nu_2 : \ker\nu \to R$ and $\mu : Rx_1 \oplus \ker\nu \to R$. For $r \in R$ and $\alpha \in \ker\nu$, we have $\mu(rx_1 + \alpha) = r\nu(x_1) + \nu_2(\alpha)$. Hence, μ is R-linear and $\mu(x) = \mu(a_1x_1) = a_1 \in \mu(N)$ so that $(a_1) = \mu(N)$. But $\mu(N) \supseteq \mu(\ker\nu \cap N) = \nu_2(\ker\nu \cap N) = (a_2)$, so $a_1 \mid a_2$. ■

April 6th.

We are now equipped to prove the structure theorem for finitely generated modules over a PID, using the previous proposition.

Proof of the structure theorem for finitely generated modules over a PID. Let $\{m_1, \dots, m_n\}$ be a generating set of M as an R-module. Then $\varphi : F = R^n \to M$ defined by $\varphi(r_1, \dots, r_n) = r_1m_1 + \cdots + r_nm_n$ for all $r_1, \dots, r_n \in R$ is a surjective R-linear map. By the previous proposition, $N = \ker(\varphi) \subseteq F$ is a free R-submodule of F and of rank $m \le n$. Moreover, there is a basis $\{x_1, \dots, x_n\}$ of F and elements $a_1, \dots, a_m \in R$ such that $\{a_1x_1, \dots, a_mx_m\}$ is a basis of N and $a_1 \mid a_2 \mid \cdots \mid a_m$. By the first isomorphism theorem, $M \cong F/N$. Note that we can write $F = Rx_1 \oplus Rx_2 \oplus \cdots \oplus Rx_n$ and $N = Ra_1x_1 \oplus Ra_2x_2 \oplus \cdots \oplus Ra_mx_m$, so

$$M \cong F/N = (Rx_1 \oplus \cdots \oplus Rx_n)/(Ra_1x_1 \oplus \cdots \oplus Ra_mx_m) \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \oplus R^{n-m}. \tag{2.7}$$

Hence, M is isomorphic to a direct sum of a free module and a torsion module. The rank of M is simply

$$\operatorname{rank} M = \dim_{S^{-1}R}(S^{-1}M) = \dim_{S^{-1}R}(S^{-1}(R^{n-m})) = n - m. \tag{2.8}$$

where $\dim_{S^{-1}R}(S^{-1}((R/(a_1)) \oplus \cdots \oplus R/(a_m))) = 0$ since $R/(a_i)$ is a torsion module for all $1 \le i \le m$. Hence, $k = n - m$. ■

There also exists an alternative version of the structure theorem for finitely generated modules over a PID.

Theorem 2.44 (The alternative structure theorem)**.** *Let R be a PID, and M be a finitely generated R-module. Then*

$$\begin{aligned} M \cong R^k &\oplus R/(p_1^{r_{11}}) \oplus R/(p_1^{r_{12}}) \oplus \cdots \oplus R/(p_1^{r_{1n_1}}) \\ &\oplus R/(p_2^{r_{21}}) \oplus R/(p_2^{r_{22}}) \oplus \cdots \oplus R/(p_2^{r_{2n_2}}) \\ &\oplus \cdots \\ &\oplus R/(p_l^{r_{l1}}) \oplus R/(p_l^{r_{l2}}) \oplus \cdots \oplus R/(p_l^{r_{ln_l}}) \end{aligned} \tag{2.9}$$

where $r_{ij} \ge r_{ij-1}$ for all $1 \le i \le l$ and $2 \le j \le n_i$, where $p_1, \dots, p_l$ are prime elements of R.

Proof. Write $M \cong R^k \oplus R/(a_1) \oplus \cdots \oplus R/(a_n)$ as in the previous structure theorem. Since R is a PID, and $a_1 \mid a_2 \mid \cdots \mid a_n$, we can write $a_i = p_1^{r_{i1}} p_2^{r_{i2}} \cdots p_l^{r_{il}}$ for some prime elements $p_1, \ldots, p_l$ of R and some non-negative integers r_{ij} such that $r_{ij} \geq r_{i,j-1}$ for all $1 \leq i \leq l$ and $2 \leq j \leq n_i$. Then, by the chinese remainder theorem, we have

$$R/(a_i) = R/(p_1^{r_{i1}} p_2^{r_{i2}} \cdots p_l^{r_{il}}) \cong R/(p_1^{r_{i1}}) \oplus R/(p_2^{r_{i2}}) \oplus \cdots \oplus R/(p_l^{r_{il}}). \tag{2.10}$$

The result follows by applying this decomposition to each $R/(a_i)$. ■

2.3.1 Applications

April 9th.

Theorem 2.45 (The *Jordan canonical form*)**.** *Let $A \in M_n(\mathbb{C})$ be an $n \times n$ complex matrix. Then there exists an invertible matrix $P \in GL_n(\mathbb{C})$ such that $P^{-1}AP = \operatorname{diag}(J_{r_1}^{\lambda_1}, \ldots, J_{r_l}^{\lambda_l})$, where $\lambda_1, \ldots, \lambda_l$ are the distinct eigenvalues of A, and*

$$J_r^\lambda = \begin{pmatrix} \lambda & 0 & \cdots & 0 \\ 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda \end{pmatrix}_{r\times r} + \begin{pmatrix} 0 & 1 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}_{r\times r} = \begin{pmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \cdots & 0 \\ 0 & 0 & \lambda & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{pmatrix}_{r\times r} \tag{2.11}$$

is the $r \times r$ Jordan block with eigenvalue λ for all $r \geq 1$ and $\lambda \in \mathbb{C}$.

Theorem 2.46 (The *rational canonical form*)**.** *Let k be a field, and $A \in M_n(k)$ be an $n \times n$ matrix over k. Then there exists an invertible matrix $P \in GL_n(k)$ such that $P^{-1}AP = \operatorname{diag}(R_{a_1}, \ldots, R_{a_m})$, where $a_1, \ldots, a_m \in k[X]$ are monic polynomials such that $a_1 \mid a_2 \mid \cdots \mid a_m$, and*

$$R_a = \begin{pmatrix} 0 & 0 & \cdots & 0 & -b_0 \\ 1 & 0 & \cdots & 0 & -b_1 \\ 0 & 1 & \cdots & 0 & -b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -b_{n-1} \end{pmatrix}_{n\times n} \tag{2.12}$$

is the companion matrix of $a(X) = X^n + b_{n-1}X^{n-1} + \cdots + b_1 X + b_0$ for all $n \geq 1$ and $a(X) \in k[X]$.

Note that if R is a ring, and M is an $R[X]$-module, then M is an R-module via the restriction of scalars, and there is a natural R-linear map $M \to M$ defined by $m \mapsto Xm$ for all $m \in M$. Conversely, suppose M is an R-module, and there is an R-linear map $\varphi : M \to M$. Then one can give M a natural $R[X]$-module structure by defining $X \cdot m = \varphi(m)$ for all $m \in M$; if $f(X) \in R[X]$, then $f(X) \cdot m$ is defined as $f(\varphi)(m)$, where $f(\varphi) : M \to M$ is the R-linear map obtained by substituting φ into $f(X)$. Hence, there is a one-to-one correspondence between $R[X]$-modules and pairs (M, φ), where M is an R-module and $\varphi : M \to M$ is an R-linear map. In particular, if we take $R = k$ to be a field, and $M = V = k^n$ to be the standard n-dimensional vector space over k, then a pair (V, A), where $A : V \to V$ is a k-linear map, corresponds to a $k[X]$-module structure on V.

Definition 2.47. Let M be an R-module, and $m \in M$. The *annihilator* of m is defined as $\operatorname{Ann}(m) = \{r \in R \mid rm = 0\}$. Note that $\operatorname{Ann}(m)$ is an ideal of R. Similarly, the annihilator of M is defined as $\operatorname{Ann}(M) = \{r \in R \mid rm = 0 \text{ for all } m \in M\} = \bigcap_{m \in M} \operatorname{Ann}(m)$, which is also an ideal of R.

Few properties may be inferred easily. For instance, M is a torsion module if and only if $\operatorname{Ann}(m) \neq 0$ for all $m \in M$. If $\operatorname{Ann}(M) \neq 0$, then M is a torsion module; converse holds if R is an integral domain and M is finitely generated. Specific to our case, let V be a finite-dimensional vector space over a field k, and φ be a k-linear map from V to itself. We already showed this gives V a natural $k[X]$-module structure. Here, $\operatorname{Ann}(v) \neq 0$. Let $p_\varphi(X)$ be the characteristic polynomial of φ. Then $(p_\varphi(X)) = \operatorname{Ann}_{k[X]}(V)$. Also, the Cayley-Hamilton theorem states that $p_\varphi(\varphi) = 0$.

April 10th.

We now prove the rational canonical form, which is a direct consequence of the structure theorem for finitely generated modules over a PID.

Proof of the rational canonical form. Let $\varphi = A : k^n \to k^n$ be the k-linear map defined by A. Since $V = k^n$ is a k-module, and φ is a k-linear map from V to itself, V is a $k[X]$-module. Let $R = k[X]$. Then R is a PID, and V is a finitely generated R-module; thus by the structure theorem for finitely generated modules over a PID, we have

$$V \cong R^{\mathrm{rank}(V)} \oplus R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_m) \tag{2.13}$$

where $\mathrm{rank}(V) = 0$ since V is a finite dimensional vector space over k, and $a_1, \ldots, a_m \in R = k[X]$ are such that $a_1 \mid a_2 \mid \cdots \mid a_m$. We may assume that the a_i's are monic since k is a field. Now define the map

$$\mu_X : R/(a_1) \oplus \cdots \oplus R/(a_m) \to R/(a_1) \oplus \cdots \oplus R/(a_m) \tag{2.14}$$

where

$$(f_1 + (a_1), \ldots, f_m + (a_m)) \mapsto (Xf_1 + (a_1), \ldots, Xf_m + (a_m)) = X \cdot (f_1 + (a_1), \ldots, f_m + (a_m)) \tag{2.15}$$

It is easy to verify that μ_X is a well-defined R-linear map. If we let θ denote the isomorphism from V to $R/(a_1) \oplus \cdots \oplus R/(a_m)$, we then claim that $\mu_X = \theta \circ \varphi \circ \theta^{-1}$. Let $v \in V$. Then

$$\varphi \circ \theta(v) = X \cdot \theta(v) = \theta(X \cdot v) = \theta \circ \varphi(v). \tag{2.16}$$

Thus, $\varphi \circ \theta = \theta \circ \varphi$, so $\mu_X = \theta \circ \varphi \circ \theta^{-1}$ as claimed. Note that $k[X]/(a_1(X))$ has basis $\{1, X, X^2, \ldots, X^{d_1-1}\}$, where $a_1(X) = X^{d_1} + b_{1,d_1-1}X^{d_1-1} + \cdots + b_{1,1}X + b_{1,0}$ by repeated division algorithm. Thus, if we let $R/(a_1)$ be the first component of $R/(a_1) \oplus \cdots \oplus R/(a_m)$, then μ_X restricted to $R/(a_1)$ is given by the companion matrix of $a_1(X)$; that is, 1 in the first component is mapped to X in the first component, X to X^2, and so on, until X^{d_1-1} is mapped to $-b_{1,0} - b_{1,1}X - \cdots - b_{1,d_1-1}X^{d_1-1}$ in the first component. So the matrix representation of μ_X with respect to the basis $\{1, X, \ldots, X^{d_1-1}\}$ of $R/(a_1)$ is the companion matrix R_{a_1} of $a_1(X)$. Similarly, the matrix representation of μ_X restricted to $R/(a_i)$ with respect to the basis $\{1, X, \ldots, X^{d_i-1}\}$ of $R/(a_i)$ is the companion matrix R_{a_i} of $a_i(X)$ for all $2 \le i \le m$. Hence, the matrix representation of μ_X with respect to the basis obtained by concatenating the bases of each $R/(a_i)$ is $\mathrm{diag}(R_{a_1}, \ldots, R_{a_m})$. Finally, if we let P be the matrix of θ with respect to the standard basis of V and the concatenated basis of $R/(a_1) \oplus \cdots \oplus R/(a_m)$, then PAP^{-1} is the matrix representation of $\mu_X = \theta \circ \varphi \circ \theta^{-1}$ with respect to the concatenated basis of $R/(a_1) \oplus \cdots \oplus R/(a_m)$, which is $\mathrm{diag}(R_{a_1}, \ldots, R_{a_m})$ as shown above. The proof is complete. ■

We now prove the Jordan canonical form, which is a direct consequence of the rational canonical form.

Proof of the Jordan canonical form. Let $k = \mathbb{C}$ and $A \in M_n(\mathbb{C})$. Note that $\varphi = A : \mathbb{C}^n \to \mathbb{C}^n$ is a $\mathbb{C}$-linear map, so $V = \mathbb{C}^n$ gains a $\mathbb{C}[X]$-module structure as before. Note that $R = \mathbb{C}[X]$ is a PID, and V is a finitely generated R-module; thus by the alternative version of the structure theorem for finitely generated modules over a PID, we have

$$\begin{aligned} V \cong & R/(p_1^{r_{11}}) \oplus R/(p_1^{r_{12}}) \oplus \cdots \oplus R/(p_1^{r_{1n_1}}) \\ & \oplus R/(p_2^{r_{21}}) \oplus R/(p_2^{r_{22}}) \oplus \cdots \oplus R/(p_2^{r_{2n_2}}) \\ & \oplus \cdots \\ & \oplus R/(p_l^{r_{l1}}) \oplus R/(p_l^{r_{l2}}) \oplus \cdots \oplus R/(p_l^{r_{ln_l}}) \end{aligned} \tag{2.17}$$

where $p_1, \ldots, p_l$ are irreducibles in $\mathbb{C}[X]$, and $r_{ij} \ge 1$. Let θ be the isomorphism from V to the above direct sum, and μ_X denote the map defined by X as before. By the fundamental theorem of algebra, we must have $p_i(X) = (X - \lambda_i)$ for some $\lambda_i \in \mathbb{C}$ for all $1 \le i \le l$. In $R/(X - \lambda_1)^{r_{11}}$, note that $\mathcal{B}_1 = \{1 + (X - \lambda_1)^{r_{11}}, (X - \lambda_1) + (X - \lambda_1)^{r_{11}}, \ldots, (X - \lambda_1)^{r_{11}-1} + (X - \lambda_1)^{r_{11}}\}$ is a basis. Observing where μ_X sends each element of $\mathcal{B}_1$, we see that the matrix representation of μ_X restricted to $R/(X - \lambda_1)^{r_{11}}$ with respect to $\mathcal{B}_1$ is the transpose of the Jordan block $J_{r_{11}}^{\lambda_1}$; writing the basis in reverse order gives the Jordan block. Similarly, the matrix representation of μ_X restricted to $R/(X - \lambda_i)^{r_{ij}}$ with respect to the basis $\mathcal{B}_{ij} = \{(X - \lambda_i)^{j-1} + (X - \lambda_i)^{r_{ij}} \mid 1 \le j \le r_{ij}\}$ is the transpose of the Jordan block $J_{r_{ij}}^{\lambda_i}$, and writing the basis in reverse order gives the required Jordan block. Hence, the matrix representation of μ_X with

respect to the basis obtained by concatenating the bases $\mathcal{B}_{ij}$ is $\operatorname{diag}(J_{r_{11}}^{\lambda_1}, \ldots, J_{r_{ln_l}}^{\lambda_l})$. Finally, if we let P be the matrix of θ with respect to the standard basis of V and the concatenated basis of the direct sum, then PAP^{-1} is the matrix representation of $\mu_X = \theta \circ \varphi \circ \theta^{-1}$ with respect to the concatenated basis, which is $\operatorname{diag}(J_{r_{11}}^{\lambda_1}, \ldots, J_{r_{ln_l}}^{\lambda_l})$ as shown above. The proof is complete. ■

Finally, we look into the Cayley-Hamilton theorem.

Theorem 2.48 (The *Cayley-Hamilton theorem*)**.** *Let R be a ring, and $A \in M_n(R)$. If $\chi_A(x) = \det(xI_n - A)$ is the characteristic polynomial of A, then $\chi_A(A) = 0$.*

Proof. Let $A = (a_{ij})_{n\times n}$, and $\varphi_A : R^n \to R^n$ be the R-linear map defined by A. Then R^n is a $R[X]$-module as before, where $f(X)\cdot v = f(\varphi_A)(v)$ for all $f(X) \in R[X]$ and $v \in R^n$. Now, $X \cdot e_j = \varphi_A(e_j) = \sum_{i=1}^n a_{ij}e_i$ for all $1 \le j \le n$, where $\{e_1, \ldots, e_n\}$ is the standard basis of R^n. Hence, $-a_{j1}e_1 - a_{j2}e_2 - \cdots - a_{j,j-1}e_{j-1} + (X - a_{jj})e_j - a_{j,j+1}e_{j+1} - \cdots - a_{jn}e_n = 0$ for all $1 \le j \le n$. Therefore,

$$XI - A = \begin{pmatrix} X - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & X - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & X - a_{nn} \end{pmatrix} \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{pmatrix} = 0. \tag{2.18}$$

Let $B = \operatorname{adj}(XI - A)$. Then $B(XI - A) = (XI - A)B = \det(XI - A)I_n = \chi_A(X)I_n$. Hence, $\chi_A(X)e_i = 0$ for all $1 \le i \le n$, or $\chi_A(X)v = 0$ for all $v \in R^n$. Thus, $\chi_A(A)v = \chi_A(\varphi_A)(v) = \chi_A(X)v = 0$ for all $v \in R^n$, so $\chi_A(A) = 0$ as an R-linear map from R^n to itself. ■

As a final note, if R_a is the companion matrix of $a(X) = X^n + b_{n-1}X^{n-1} + \cdots + b_1X + b_0$, then $\chi_{R_a}(X) = a(X)$, so the Cayley-Hamilton theorem implies that $a(R_a) = 0$. Hence, in the rational canonical form, for the matrix A which is similar to $\operatorname{diag}(R_{a_1}, \ldots, R_{a_m})$, we have $\chi_A(X) = a_1(X)a_2(X)\cdots a_m(X)$. Moreover, A must have minimal polynomial $p_A(X) = a_m(X)$. Lastly, if for such a matrix A we have $\chi_A(X) = p_A(X)$, then $P^{-1}AP = R_{\chi_A(X)}$.

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